

# Haircutting Robots: The Arrival of a New Scientific Frontier

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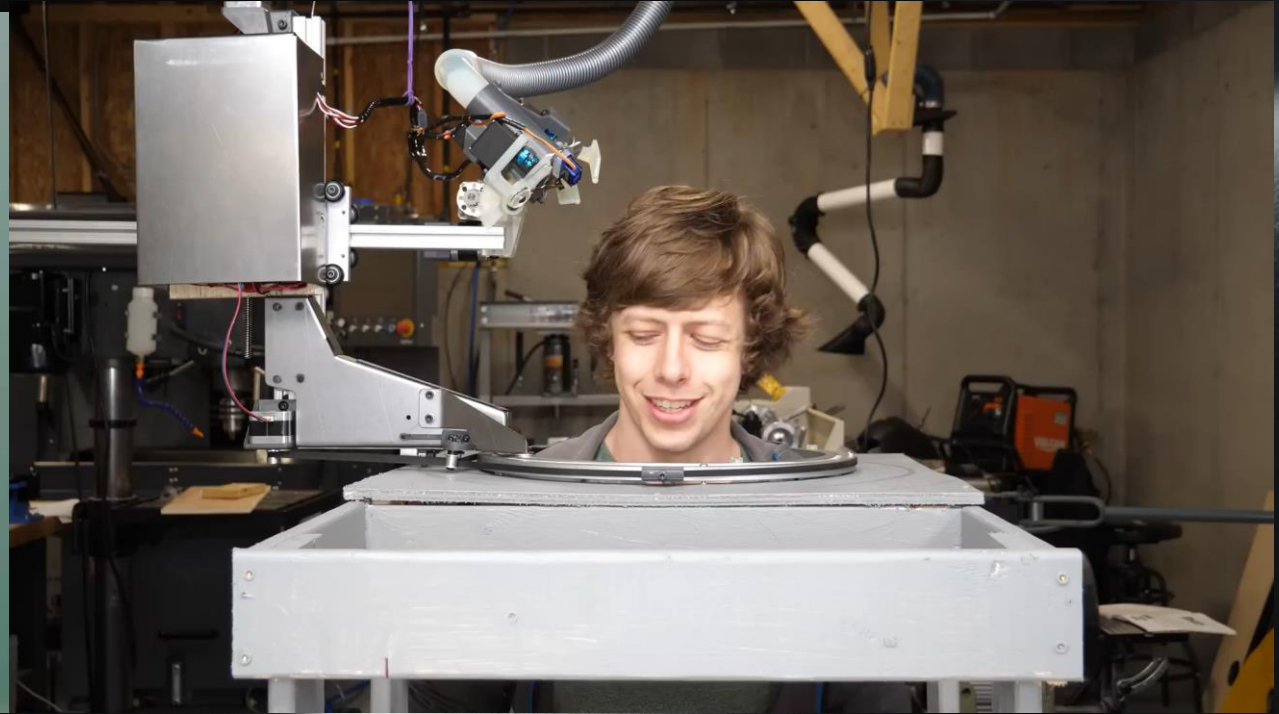


The background is a dark, deep space scene. In the upper left, a portion of a planet with a ring system is visible. In the lower right, a large, bright, curved horizon of a planet dominates the frame. The background is filled with numerous small, distant stars.

01

# Introduction & Motivation

# Prototypes on Haircutting Robots



# Prototypes on Haircutting Robots



**STEP 1: 3D scan the neck using iPhoneX**

# Prototypes on Haircutting Robots



# Imaginary Haircutting Robots



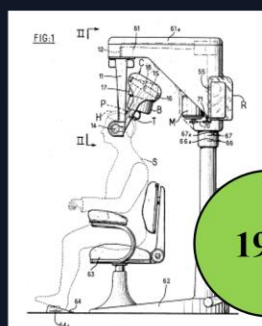
Robot ने काटे  
Elon Musk के hair

# Imaginary Hair cutting/dressing Robots



# Brief History of Haircutting Robots

## The first patent on haircutting



1961

## United States Patent Office

3,241,562  
AUTOMATIC HAIR-CUTTING MACHINE HAVING  
PROGRAMMED CONTROL MEANS FOR CUT-  
TING HAIR IN A PREDETERMINED STYLE  
Jean Gronier, 145 Blvd. de la Reine, Versailles, France  
Filed Feb. 10, 1961, Ser. No. 88,409  
Claims priority, application France, Feb. 16, 1960,  
818,578; Feb. 1, 1961, 851,348  
13 Claims. (Cl. 132-45)

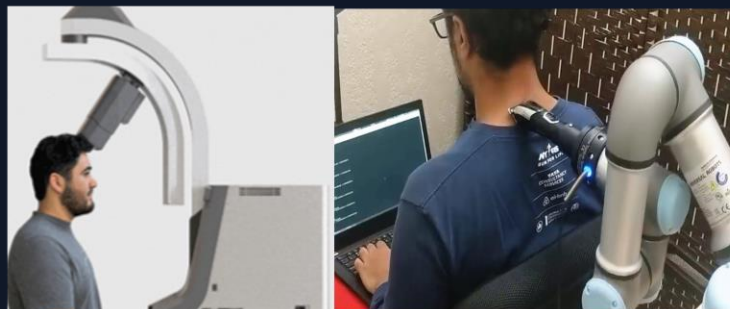
My invention has for its object a hair-cutting machine comprising means for the automatic cutting of hair in accordance with a predetermined program; each program is designed for a particular person and is established once and for all with a view to obtaining repeatedly the same cut for the same head. More specifically, the machine includes a frame, implements for cutting the hair, movable means connecting the instruments with the frame,

from above the array third change speed go FIG. 19 is a pers 18 of the same part FIG. 20 is a pers the connection bet and the casing carry FIGS. 21, 22, 23 third change speed ferent shafts formi FIG. 25 is a view the casing carrying in the direction of FIG. 26 is a dia gear system contri towards or away fro FIG. 27 is a long ment carrier.



Covid

## A burst of DIY robotic haircutting



## Conference special issue on haircutting robots

Special Session – Haircutting Robots: Intelligent Sensing, Dexterous Control, and Human-Centric Design

### Scope and Objectives

As robotics expands into service industries, haircutting emerges as a uniquely challenging domain requiring human-like dexterity, personalization, and ergonomic consideration. These systems must navigate complex head geometries while ensuring user comfort, adapting to individual anthropometrics and maintaining safety in dynamic environments. Traditional robotic solutions, optimized for structured tasks, struggle with these contact-rich, ergonomically sensitive operations that demand adaptive precision and human-centric interaction.

### Submission Guidelines

Papers submitted to this special session should follow the standard formatting and submission guidelines of ICNSC 2025 which can be found on the Submission page: <https://www.cdu.edu.au/icscsc2025/submission.html>.

During the submission process, authors should select this special session by title (Special Session on Haircutting Robots: Intelligent Sensing, Dexterous Control, and Human-Centric Design) to ensure their papers are directed appropriately.

### Session Organizer

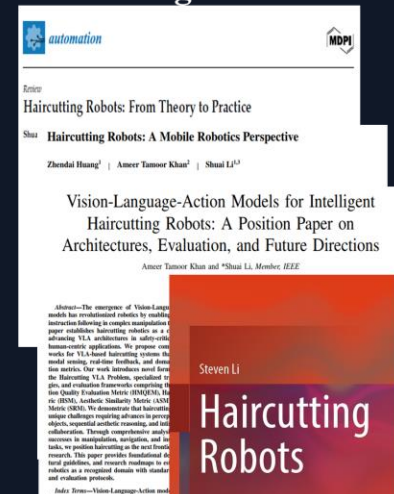
Session Organizer: Dr. Ata Jahangir Mohtashami, Associate Professor

Affiliation: School of

Department of

2025

## Scientific publications on haircutting robots



AI Video Generation

## A burst of AI-generated videos on robotic haircutting



Keynote speech: IEEE 25th International Symposium on Computational Intelligence and Informatics (CINTI 2025) on November 18–20, 2025 in Budapest, Hungary.  
Keynote speech: IEEE 2nd International Conference on Advanced Computing and Emerging Technologies (ACET 2025) on November 21, 2025.

# Why Haircutting Robots Matter



## Market Drivers

- Rising labor costs & barber shortages
- Demand for consistent, high-quality service
- 24/7 availability and enhanced hygiene



## Robotic Solutions

- Create a new sector in automation
- Offer personalized, efficient experiences
- Address a massive, recurring market need

# Rise of Affordable Robots



Then (Honda ASIMO)

~\$1.3M

Now (Unitree R1)

~\$5,900

This ~200x cost reduction unlocks new service sectors like haircutting, previously economically unfeasible.

The background is a dark, cosmic scene. In the upper left, a portion of a planet with rings is visible. In the lower right, a large, bright, curved horizon of a planet dominates the frame. The background is filled with numerous small, distant stars.

02

# Market Opportunity

# Urban Haircut Spending

Annual spending in major cities demonstrates strong local demand for automated solutions.

NYC annual market size:  $4.125 \text{ million men} \times 12 \times 20\text{USD} \times 2 = 1.98 \text{ billion USD}$

~2.0

New York City  
Billion USD

~9.0

Shanghai  
Billion RMB

~3.5

London  
Billion GBP

# Global Market Estimate

Segmenting the global population reveals a massive market opportunity.

**\$576B**

High-Income  
1.2B People

**\$216B**

Middle-Income  
2.0B People

**\$35B**

Low-Income  
1.45B People

Total Annual Global Market: ~\$827 Billion USD

Ref. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation

A dark space-themed background featuring a large, bright, curved horizon of a planet or moon in the lower right. In the upper left, a ringed planet like Saturn is partially visible. The background is filled with numerous small, distant stars.

03

Prior Research

# Computer Graphics Foundations

Table 1. Existing works on hair modeling and hair editing.

| References | Tasks                                                       | Application to Haircutting Robots | Code Availability                                                                                             |
|------------|-------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------|
| [3]        | Hair simulation                                             | Hair preview                      | No                                                                                                            |
| [4]        | Hair 3D reconstruction                                      | 2D image to hair 3D model         | <a href="https://github.com/hoseok-tong/NeuralHaircutTHS">https://github.com/hoseok-tong/NeuralHaircutTHS</a> |
| [5]        | Hair segmentation                                           | Hair preview and recommendation   | No                                                                                                            |
| [6]        | Hair 3D reconstruction from images                          | 2D image to hair 3D model         | <a href="https://github.com/papagina/Hair-Net">https://github.com/papagina/Hair-Net</a> DataSetGeneration     |
| [7]        | Hair 3D reconstruction from images                          | 2D image to hair 3D model         | No                                                                                                            |
| [8]        | CT based hair 3D reconstruction                             | CT based hair 3D model            | <a href="https://github.com/facebookresearch/CT2Hair">https://github.com/facebookresearch/CT2Hair</a>         |
| [9]        | Hair dynamic simulation                                     | Hair preview                      | No                                                                                                            |
| [11]       | Image based hair style and color editing                    | Hair preview and recommendation   | <a href="https://github.com/Zlin0530/Hair-Manip">https://github.com/Zlin0530/Hair-Manip</a>                   |
| [12]       | Strand-based 3D hair model and hair style and color editing | Hair preview and recommendation   | No                                                                                                            |
| [13]       | Video based dynamic image manipulation and editing          | Hair preview and recommendation   | No                                                                                                            |

**Physically-based Strand Models**

Simulate hair's dynamic behavior.

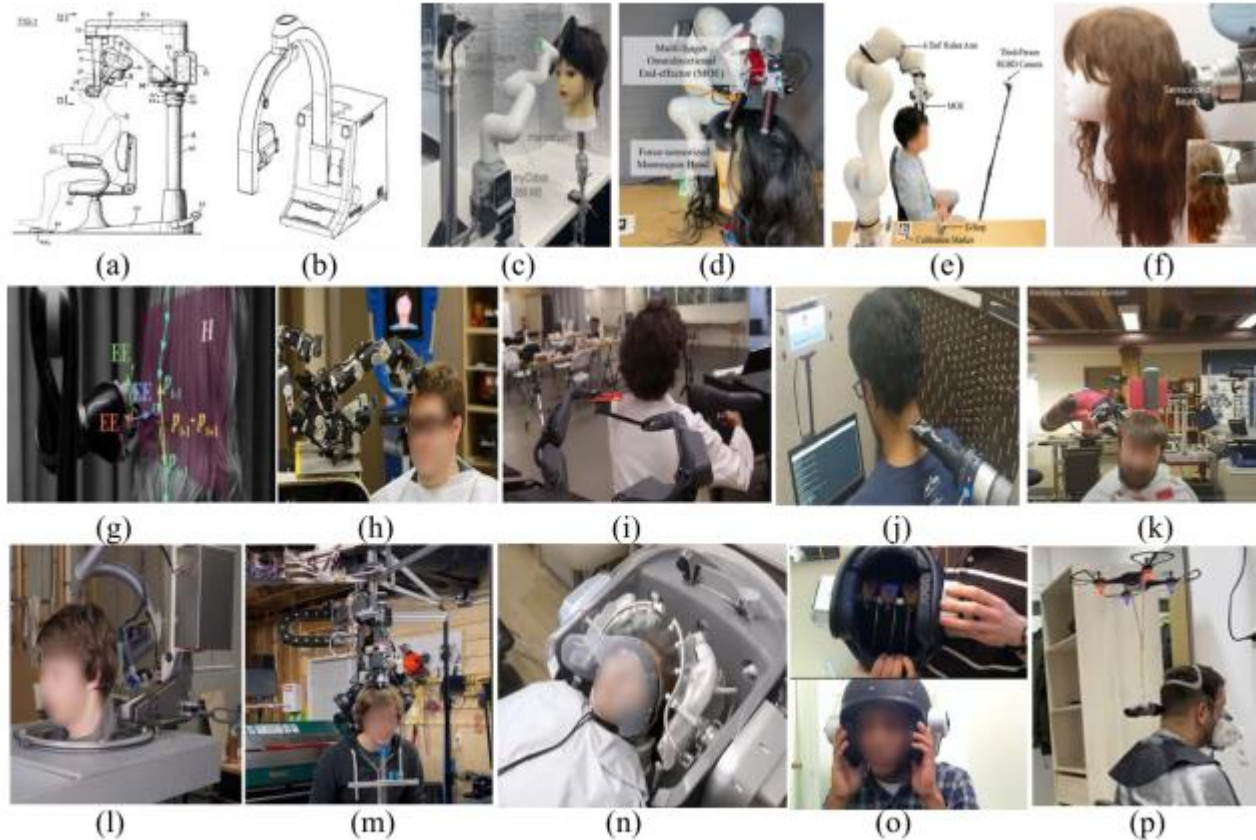
**Single-View 3D Reconstruction**

Create accurate hair models from images.

**Dynamic Simulation**

Enable realistic style preview.

# Robotic Hair Manipulation



**Figure. 1.** Sixty-year of endeavor in human history towards haircutting robots [14,16-30]. (descriptions summarized in Table 1).

**Table 2.** Existing attempts towards haircutting robots.

| References | No. in Fig. 1 | Publication format | Type               | Targeted area | Description                                                              |
|------------|---------------|--------------------|--------------------|---------------|--------------------------------------------------------------------------|
| [14]       | a             | Patent             | System development | Product       | Mechanical and circuit design                                            |
| [16]       | f             | Paper              | Programming        | Research      | Coding on commercial robots with force feedback                          |
| [17]       | e             | Paper              | Programming        | Research      | Coding on commercial robots with force feedback                          |
| [18]       | d             | Paper              | System development | Research      | Coding on self-developed soft gripper with commercial robots             |
| [19]       | c             | Paper              | Programming        | Research      | Coding on commercial robots for front hair styling                       |
| [20]       | g             | Paper              | Programming        | Research      | Coding on commercial robots for vision-based hair combing                |
| [21]       | n             | Paper              | Research           | Product       | A commercial product by Panasonic Ltd for shampooing and scalp massaging |
| [22]       | b             | Patent             | System development | Product       | A commercial product by StudioRed Ltd for MyBarberRobot.                 |
| [23]       | l             | Video              | System development | DIY           | A vision-based DIY system for automated haircutting                      |
| [24]       | m             | Video              | System development | DIY           | A vision-based DIY system for automated haircutting                      |
| [25]       | j             | Video              | Programming        | DIY           | iPhone scanning based automated haircutting using a commercial robot     |
| [26]       | k             | Video              | Programming        | DIY           | Automated haircutting using a collaborative robot                        |
| [27]       | h             | Video              | Programming        | DIY           | Teleoperated haircutting                                                 |
| [28]       | i             | Video              | Programming        | DIY           | Teleoperated haircutting                                                 |
| [29]       | o             | Video              | Future             | DIY           | Imaginary haircutting helmet                                             |
| [30]       | p             | Video              | Future             | DIY           | Conceptual design with drones for haircutting                            |

Refs. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation;  
S. Li, Book: Haircutting Robots, Springer.

# Robotic Hair Manipulation

## Online hair simulation/manipulation tools

| Name          | Application       | Type        | Website                                                                                                                                                               |
|---------------|-------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sisar         | Hair grooming     | Software    | <a href="https://sisir.sisir-hairtool.workers.dev/">https://sisir.sisir-hairtool.workers.dev/</a>                                                                     |
| Blender       | Hair simulation   | Software    | <a href="https://docs.blender.org/manual/en/latest/physics/particles/hair/index.html">https://docs.blender.org/manual/en/latest/physics/particles/hair/index.html</a> |
| Digital Salon | Hair grooming     | Open source | <a href="https://github.com/digital-salon/Digital-Salon">https://github.com/digital-salon/Digital-Salon</a>                                                           |
| HairCombSim   | Hair combing      | Open source | <a href="https://github.com/chengyzzhao/HairCombSim">https://github.com/chengyzzhao/HairCombSim</a>                                                                   |
| Dymohair      | Hair Manipulation | Open source | <a href="https://dymohair.github.io/">https://dymohair.github.io/</a>                                                                                                 |
| Tressfx       | Hair simulation   | AMD library | <a href="https://gpuopen.com/tressfx/">https://gpuopen.com/tressfx/</a>                                                                                               |
| Tfhair        | Hair simulation   | Open source | <a href="https://github.com/xinyazhang/tfhair">https://github.com/xinyazhang/tfhair</a>                                                                               |

A dark space-themed background featuring a large, bright, curved horizon of a planet or moon in the lower right. In the upper left, a ringed planet like Saturn is partially visible. The background is filled with numerous small, distant stars.

04

# Technical Requirements

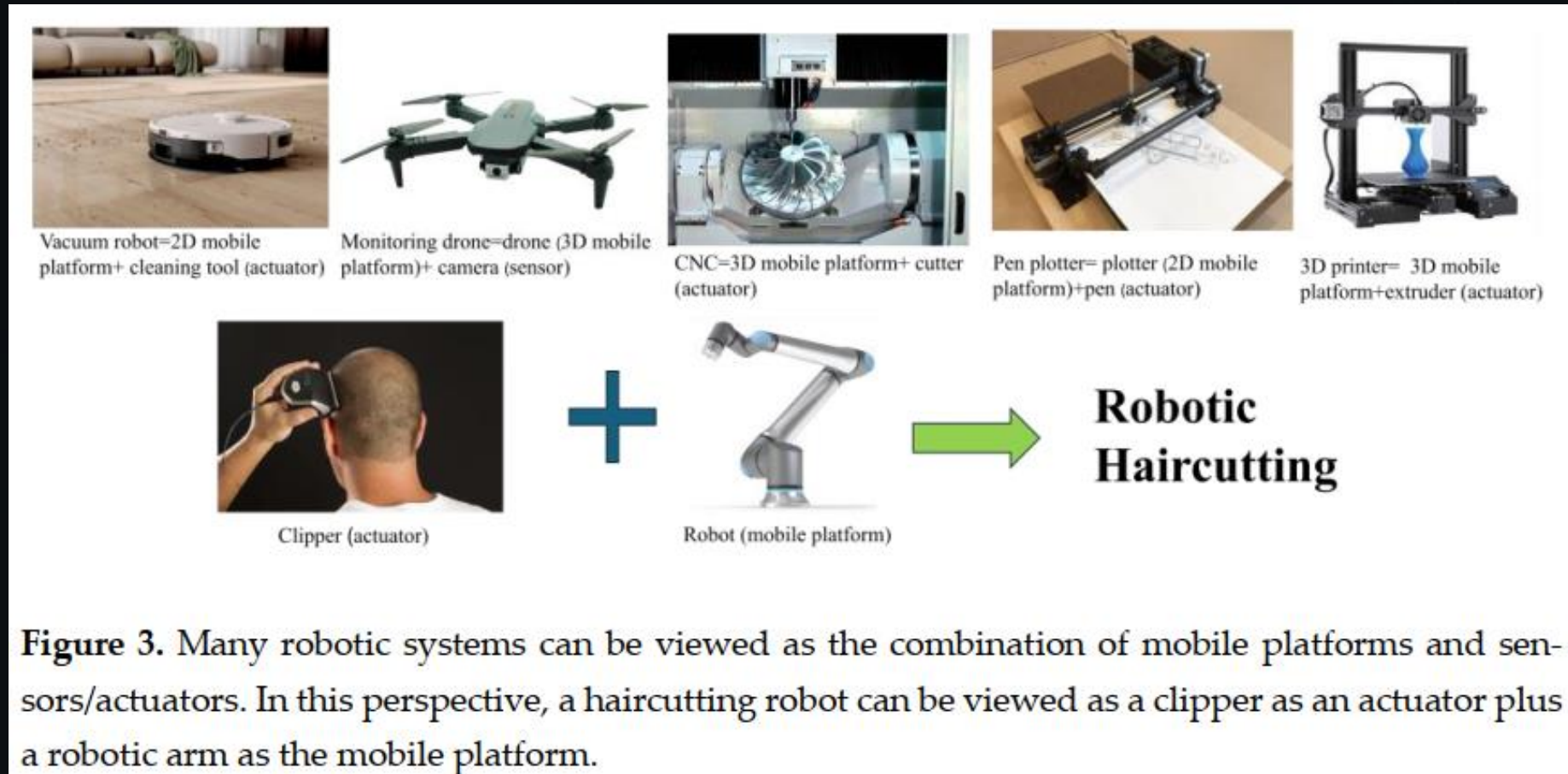
# Teleoperated Cutting Needs

Remote operation must overcome fundamental **communication delays** to ensure stability and safety.



**Figure 2.** Teleoperated haircutting, which allows remote haircut of customers in metropolitan areas by barbers living in rural areas. (This figure is partially edited with Gemini 2.5 Flash).

# Autonomous Haircutting



Refs. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation

A. Khan and S. Li Robotic Haircutting Systems: A Survey of Methods, Challenges, and Hair Modeling Insights, IEEE Journal of Selected Areas in Sensors.

# Autonomous Cutting Pipeline



Clipper (actuator)



Robot (mobile platform)



**Robotic  
Haircutting**

UFO Clipper

## EVEN CUT™

Conair EVEN CUT is sold as  
**BaByliss<sup>®</sup> men EASY CUT**  
in Europe.

Products and operation  
are identical, so watch  
our BaByliss Easy Cut  
video to see how the  
Conair EVEN CUT works.

**BaByliss<sup>®</sup>**

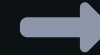


**CONAIR**

Flowbee Clipper



# Autonomous Cutting Pipeline: A CNC Perspective



## 1. Workpiece Definition

Treat head as a dynamic 3D object.

## 2. 5-Axis Toolpath Planning

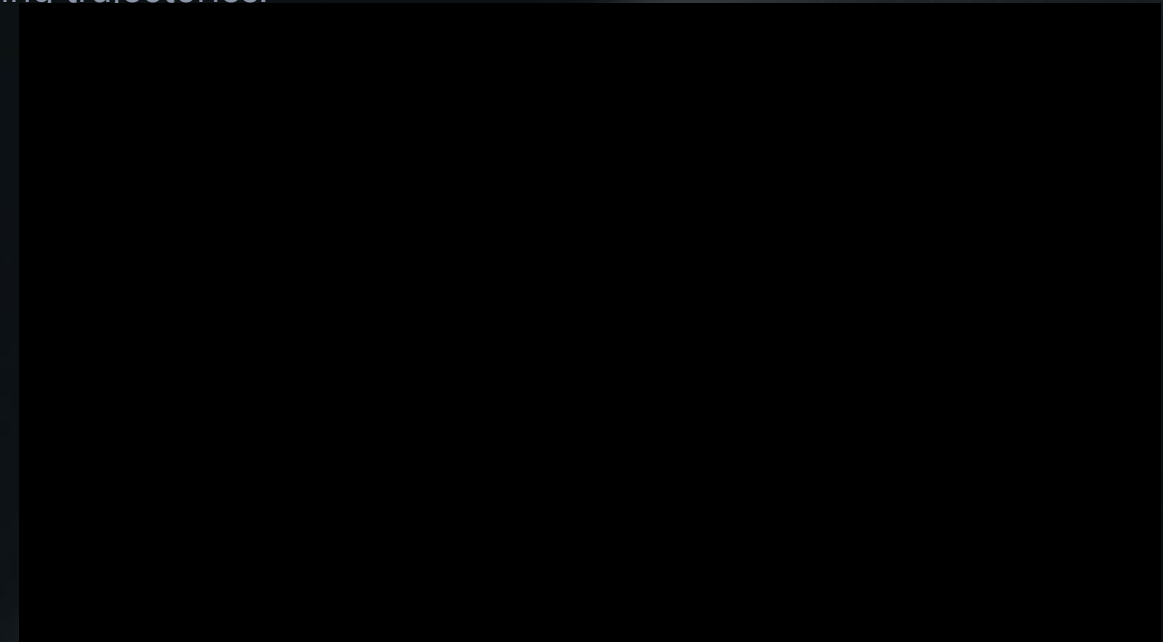
Generate precise cutting trajectories.

## 3. Real-Time Tracking

Adapt to head motion with visual servoing.

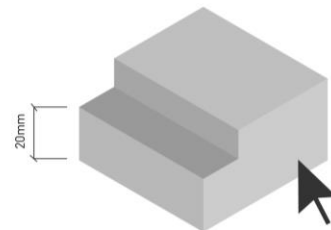
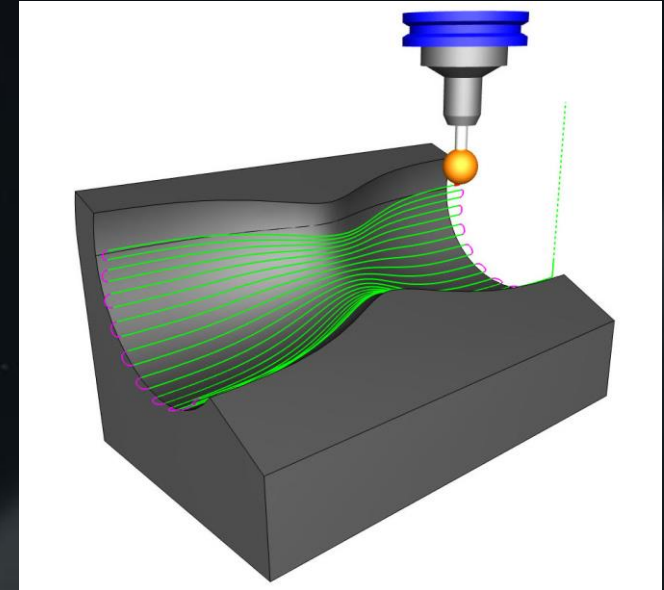
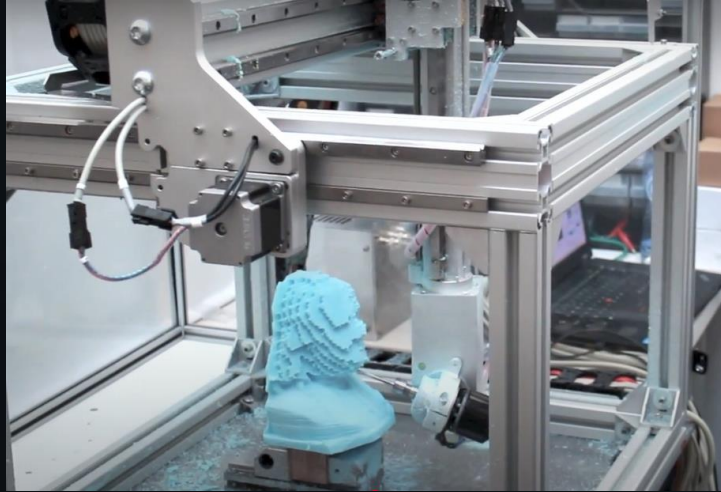


CNC for Sculpture

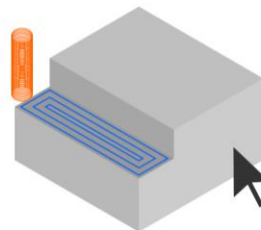


KUKA robot for wood carving

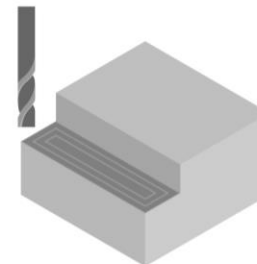
# CNC Perspective



CAD  
(Computer Aided Design)



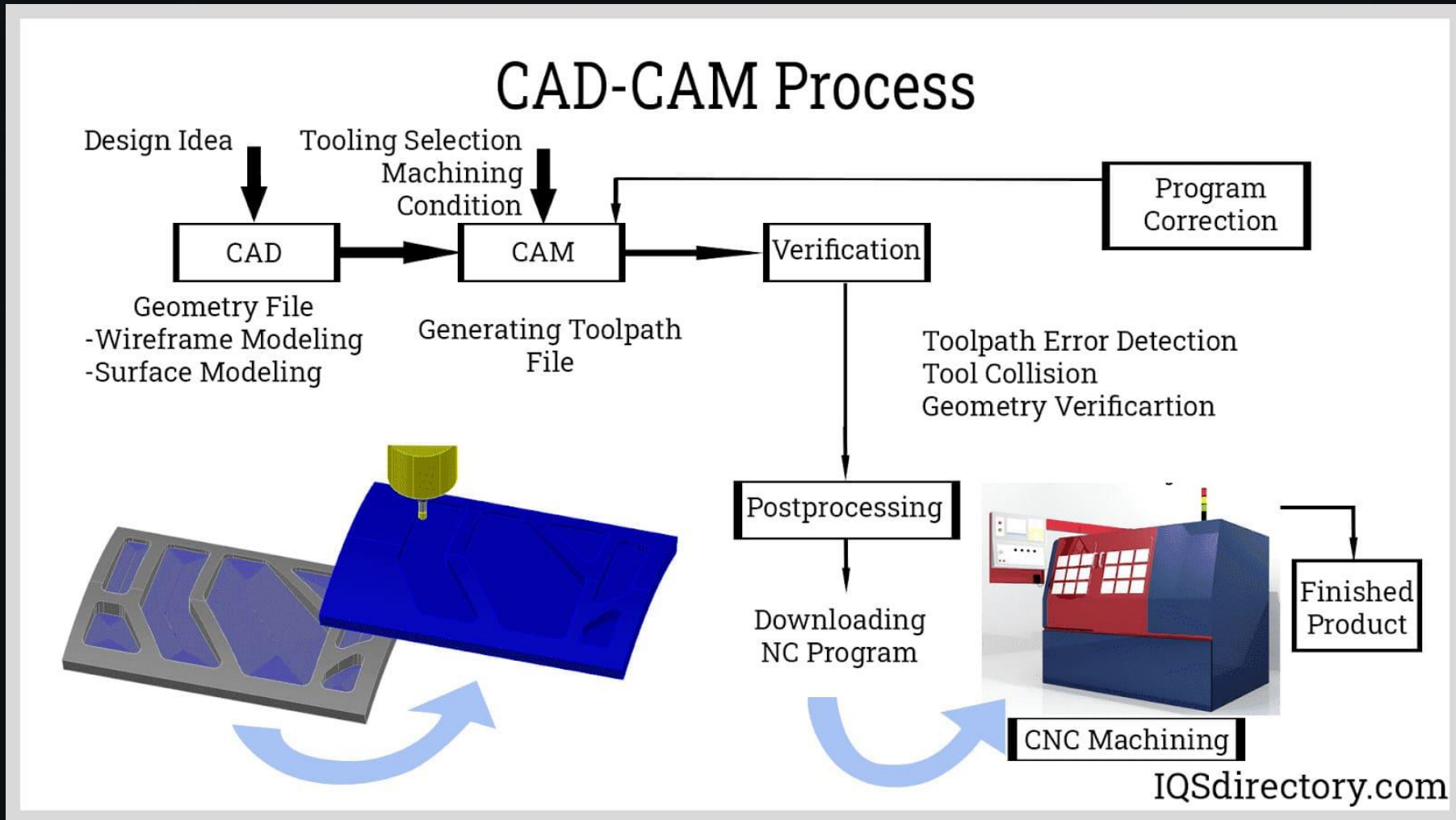
CAM  
(Computer Aided Manufacture)



CNC  
(Computer Numerical Control)

Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

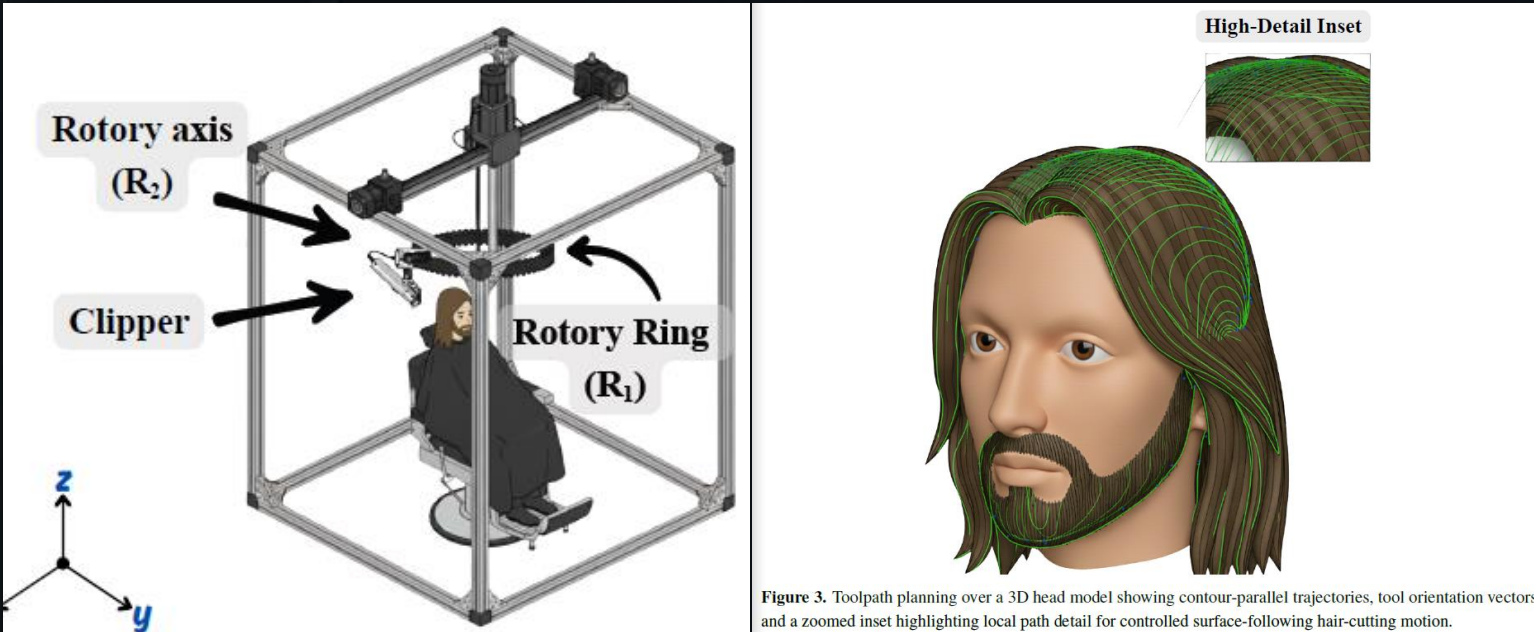
# CNC Perspective



<https://www.iqsdirectory.com/articles/cnc-machining.html>

Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

# CNC Perspective



## Algorithm 1 Multi-Pass Toolpath Generation

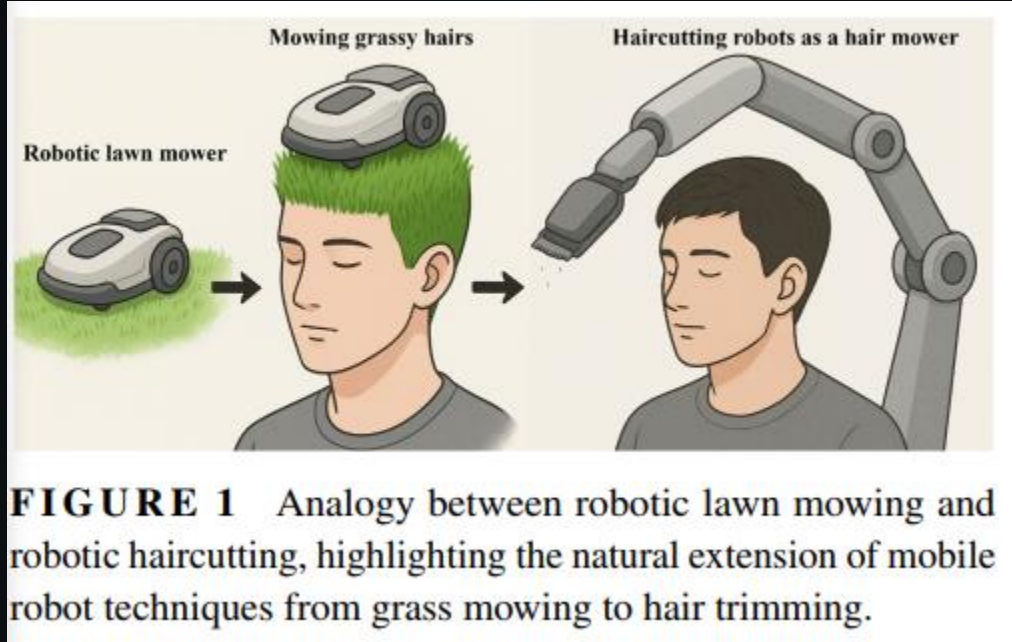
**Require:** Initial hair volume  $S_0$ , Target hairstyle  $T$ , Safety zones  $O$

**Ensure:** Sequence of toolpaths  $\{g_i(t)\}_{i=1}^N$

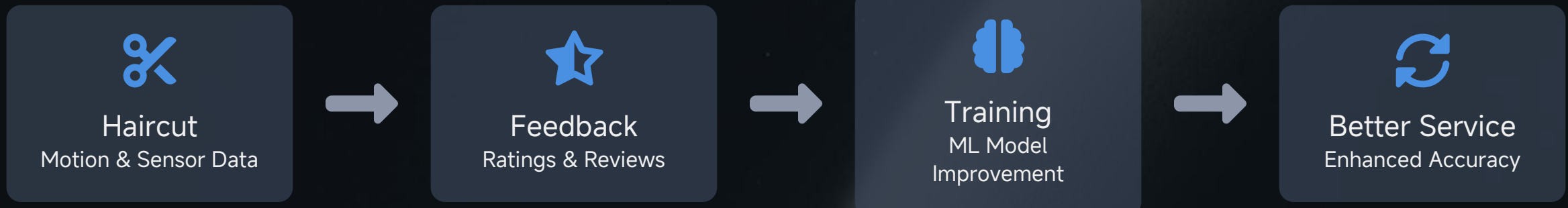
- 1: Initialize residual volume  $S \leftarrow S_0$
- 2:  $i \leftarrow 0$
- 3: **while**  $S \not\subseteq T^{+\epsilon}$  **do**
- 4:    $i \leftarrow i + 1$
- 5:   Compute removal target  $R_i \leftarrow S \setminus T$
- 6:   Generate collision-free toolpath  $g_i(t)$  covering  $R_i$
- 7:   Verify  $g_i(t) \cap O = \emptyset$  for all  $t$
- 8:   Execute pass and update  $S \leftarrow S \setminus E_i$
- 9: **end while**
- 10: **return**  $\{g_i(t)\}_{i=1}^N$

Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

# Mobile Robotics Perspective



Continuous data collection creates a virtuous cycle of improvement.



# Mobile Robotics Perspective

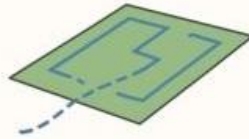
Mobile  
Robot

Where am I?



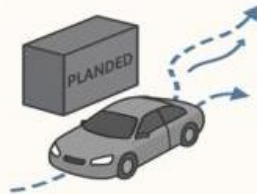
Robot vs Environment

Where to go?



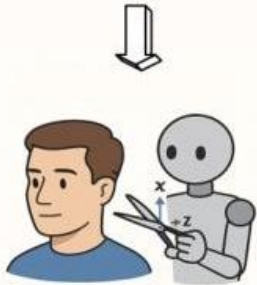
Cover entire area

How can I go there?

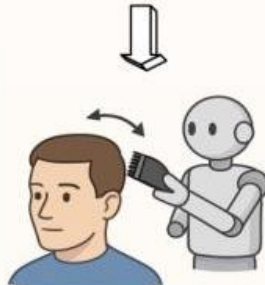


Efficiency & Corridor

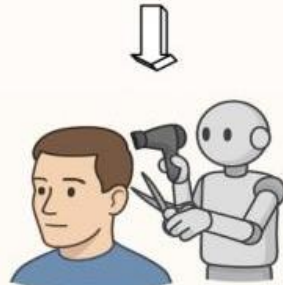
Haircutting  
Robot



Tools vs Head



Cover entire area  
with hairstyle



Safety & Coordination



## Localization: "Where am I?"

Equates to **Tool-Head Pose Estimation**: Accurately tracking the 3D position and orientation of scissors/clippers relative to the human scalp in real time.



## Coverage Planning: "Where to go?"

Equates to **Trimming Trajectory Generation**: Designing collision-free, full-coverage paths to sculpt the initial hair volume into the desired hairstyle geometry.



## Control & Execution: "How to get there?"

Equates to **Safe Motion Execution**: Executing the plan with compliant force control to adapt to head movement and ensure human safety.

# Mobile Robotics Perspective

## Key Definitions

**Definition 1** ((Effect of clipper pose)). Let  $K \subset \mathbb{R}^3$  denote the geometric model of the clipper in its local coordinate frame. During the  $i$ -th trimming pass, the pose of the clipper  $g_i(t)$  at time  $t \in I_i$ , where  $I_i \subset \mathbb{R}$  denotes the time interval of the  $i$ -th trimming pass, is described by a rigid-body transformation:

$$g_i(t) = (R_i(t), p_i(t)) \in SE(3),$$

where  $R_i(t) \in SO(3)$  is the rotation matrix and  $p_i(t) \in \mathbb{R}^3$  is the translation vector. The region occupied by the clipper in the global coordinate system is defined as:

$$g_i(t) \otimes K := \{R_i(t)x + p_i(t) \mid x \in K\}. \quad (1)$$

**Definition 2** ((Swept volume of the  $i$ -th pass)). The swept volume of the  $i$ -th trimming pass, denoted by  $\mathcal{W}_i$ , is defined as the union of all occupied regions throughout the motion interval:

$$\mathcal{W}_i := \bigcup_{t \in I_i} g_i(t) \otimes K \subset \mathbb{R}^3. \quad (2)$$

**Definition 3** ((Effective removal volume and residual volume)). Let  $S_0 \subset \mathbb{R}^3$  be the initial uncut hair volume. For each pass  $i = 1, 2, \dots, N$ , the effective removal volume  $E_i$  and the residual hair volume  $S_i$  are defined as

$$\begin{cases} E_i := (\mathcal{W}_i \cap S_{i-1}) \\ S_i := S_{i-1} \setminus E_i, \end{cases} \quad (3)$$

where  $E_i$  is the portion of hair actually cut during pass  $i$ , and  $S_i$  is the hair that remains afterwards.

**Definition 4** ((Total trimmed volume)). The total trimmed volume after  $N$  passes, denoted by  $R_N$ , is defined as:

$$R_N := \bigcup_{i=1}^N E_i = S_0 \setminus S_N. \quad (4)$$

# Mobile Robotics Perspective

## Constraints

### Trimming Depth Constraint

At each pass, the maximum amount of hair removed is limited by

$$\text{depth}(x, S_{i-1}) := \inf_{y \in \partial S_{i-1}} \|x - y\| \leq d_{\max}, \quad \forall x \in E_i, \quad (5)$$

where  $d_{\max} > 0$  specifies the allowable cutting depth relative to the pre-trimmed surface  $\partial S_{i-1}$ .

## Related Definitions

**Single-pass trimming** ( $d_{\max} = \infty$ )

**Multi-pass trimming** ( $d_{\max} < \infty$ )

**Coarse trimming:** Quickly removes a large portion of hair to approximate the rough silhouette of the target hairstyle;

**Fine finishing:** Refines edges, transitions, and surface smoothness to meet aesthetic requirements.

### Collision Avoidance Constraint

Let  $O \subset \mathbb{R}^3$

represent the obstacle volume, including anatomical structures (ears, eyes, nose), restricted scalp regions, and previously trimmed surfaces. At time  $t \in I_i$ , the occupied volume of the clipper system is

$$K_{\text{full}}(t) := g_i(t) \otimes K_{\text{tool}} \cup \mathcal{L}(g_i(t)), \quad (6)$$

where  $K_{\text{tool}} \subset \mathbb{R}^3$  denotes the clipper geometry and  $\mathcal{L}(g_i(t))$  the spatial extent of all attached mechanical links under pose  $g_i(t)$ . The non-penetration requirement is

$$K_{\text{full}}(t) \cap O = \emptyset, \quad \forall t \in I_i. \quad (7)$$

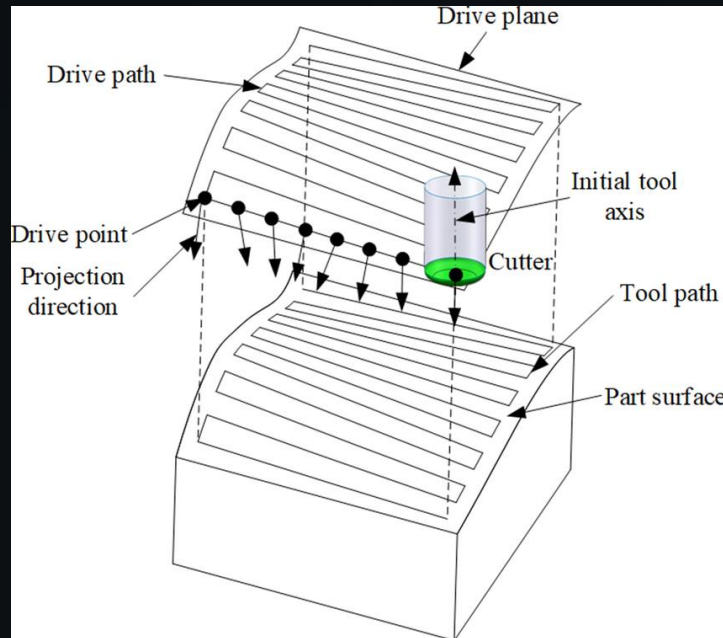
# Mobile Robotics Perspective

## Haircutting Problem Definition

Trimming Depth

Obstacle Avoidance

Desired Hair Shape



Toolpath for workpiece machining

**Definition 7** ((Feasible haircutting coverage planning problem)). The planning task is to find a sequence of trajectories  $\{g_i(t)\}_{i=1}^N$  such that the final residual hair satisfies

$$T^{-\varepsilon} \subseteq S_N \subseteq T^{+\varepsilon}, \quad (8)$$

while obeying the trimming depth and collision avoidance constraints. Here  $T \subset \mathbb{R}^3$  is the target hairstyle, and  $T^{+\varepsilon}, T^{-\varepsilon}$  denote its morphological dilation and erosion with tolerance margin  $\varepsilon > 0$ .

## Optimality Index

**Time-optimal execution:** Complete the trimming task in the shortest time, minimizing delays from velocity limits, direction changes, or reconfiguration.

**Energy-efficient execution:** Minimize energy consumption, approximated by cumulative actuation effort or total path length, subject to feasibility.

The background is a dark, cosmic scene. On the right side, a bright, curved horizon of a planet or moon is visible, glowing with a soft light. The rest of the background is black, filled with numerous small, white stars of varying brightness. In the top left corner, there is a faint, dark, curved shape that might be a satellite or a distant planet.

05

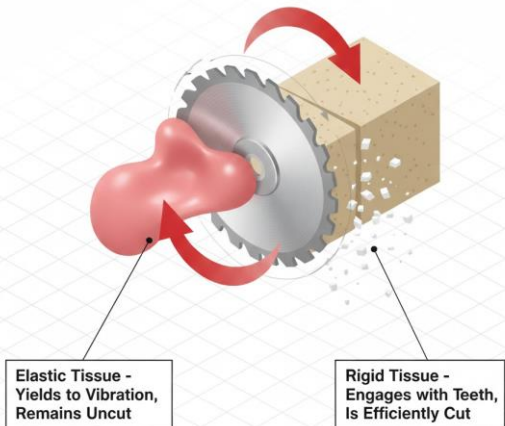
Safety & Security

# Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

✂ Sharp Tools: Use safer alternatives like clippers.

Selective Cutting Principle



# Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

TABLE 2 Categorization of risks in haircutting robots across system aspects

| Risk category           | Typical issues                                                                                                                                                                                                                                                                                                                                                | Associated components                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Network & cybersecurity | <ul style="list-style-type: none"><li>• High latency in teleoperation causing delayed reaction</li><li>• Network congestion leading to packet loss</li><li>• Cyberattacks that hijack robot control</li><li>• Unauthorized modification of haircut trajectory</li><li>• Service interruption from unstable wireless connection</li></ul>                      | Communication link, wireless modules, teleoperation interface, network protocols                    |
| Perception & sensing    | <ul style="list-style-type: none"><li>• Misclassification of hair, skin, or ear boundaries</li><li>• Depth camera errors due to lighting or occlusion</li><li>• Force/torque sensor drift or calibration failure</li><li>• Loss of vibration or acoustic signals during tool–hair interaction</li><li>• Single-sensor dependency without redundancy</li></ul> | Cameras, depth sensors, force/torque sensors, acoustic/vibration sensors, sensor fusion modules     |
| Physical & tool-related | <ul style="list-style-type: none"><li>• Excessive contact force from clippers or scissors</li><li>• Unsafe blade geometry causing skin injury</li><li>• Overheating of dryers or curlers near scalp</li><li>• Accidental collision with robot arm or end-effector</li><li>• Tool wear leading to irregular performance</li></ul>                              | Clippers, scissors, heating tools (dryer/curler), blade geometry, robotic arm end-effector          |
| Control & algorithmic   | <ul style="list-style-type: none"><li>• Instability of controller under time delay</li><li>• Unsafe motion planning without safety margins</li><li>• Lack of fallback strategies when sensors fail</li><li>• Runtime monitoring failure to stop hazardous action</li><li>• Inconsistent tool switching logic</li></ul>                                        | Motion planner, obstacle avoidance module, controller, runtime safety monitor, tool-switching logic |
| Human-related           | <ul style="list-style-type: none"><li>• Sudden head movements (rotation, tilting, reflexes)</li><li>• Postural shifts due to discomfort or fatigue</li><li>• Fear or psychological stress during close contact</li><li>• Non-cooperation or unexpected user actions</li><li>• Lack of trust or acceptance of robotic haircutting</li></ul>                    | User head, posture, facial regions (ear/neck), behavioral response, psychological state             |



Sharp Tools: Use safer alternatives like clippers.



Excessive Force: Implement soft, compliant materials.



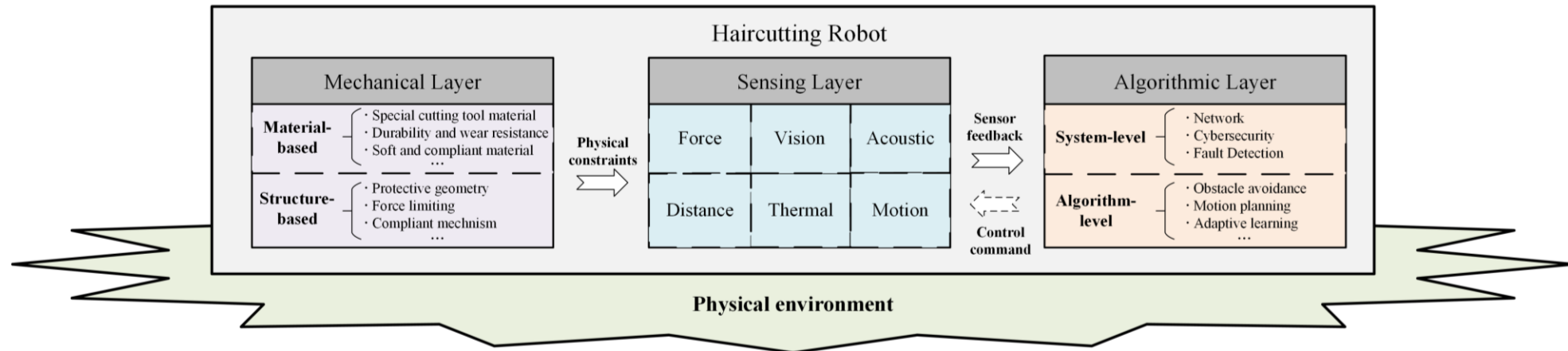
Sensor Failure: Use redundant sensor systems.



Cybersecurity: Employ local safety monitors.

# Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.



**FIGURE 3** Three-layer safety framework for haircutting robots with layer-wise components



Sharp Tools: Use safer alternatives like clippers.



Sensor Failure: Use redundant sensor systems.



Excessive Force: Implement soft, compliant materials.



Cybersecurity: Employ local safety monitors.

# Security Certification

## Supervised Autonomy Model



Cybersecurity: Employ local safety monitors.

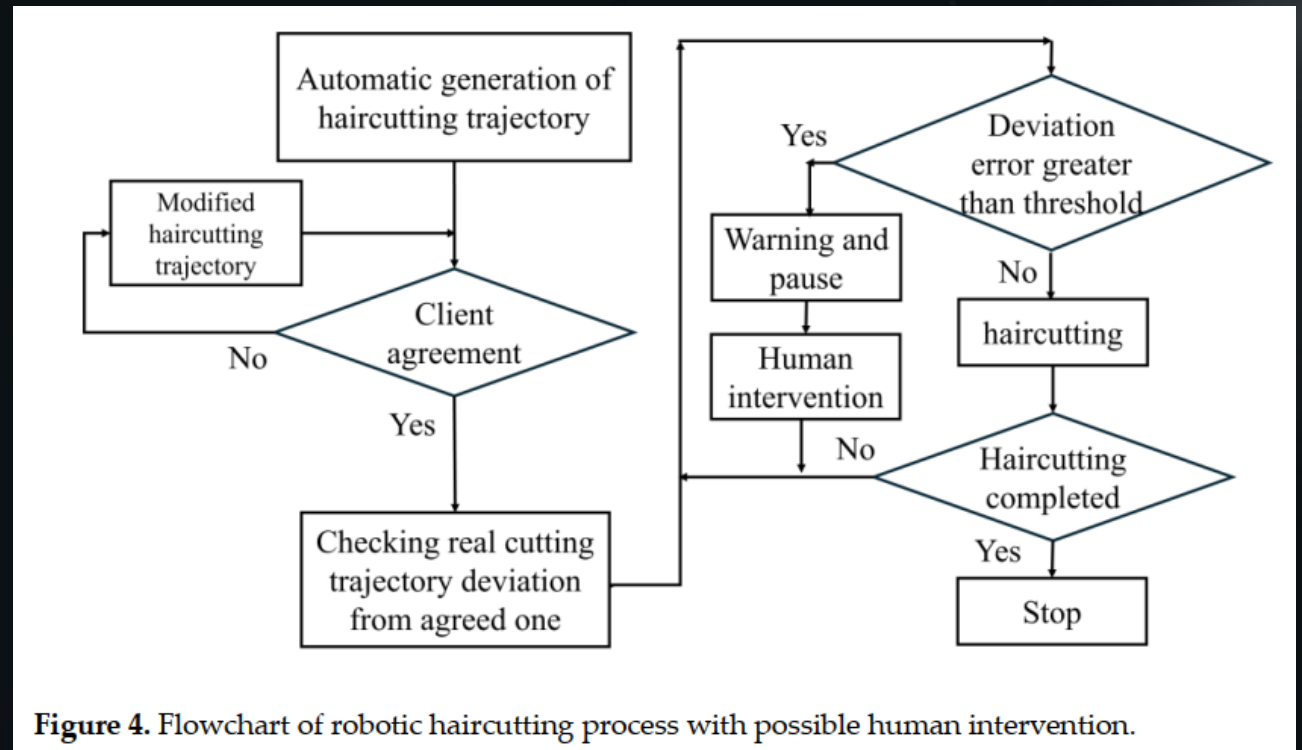
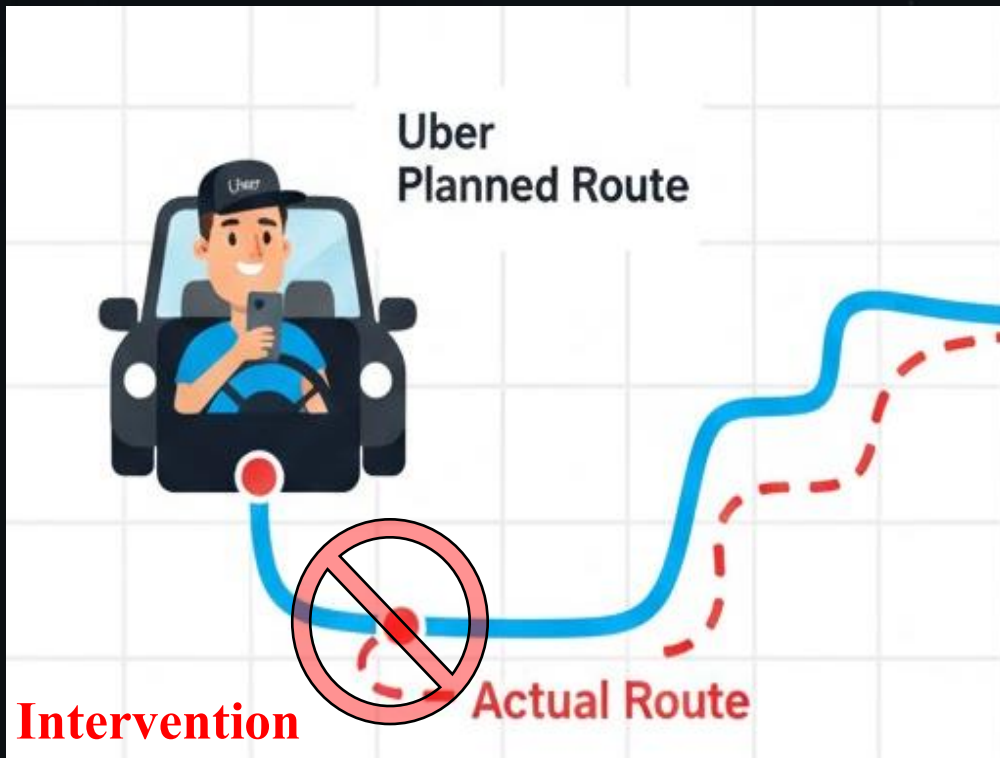


Figure 4. Flowchart of robotic haircutting process with possible human intervention.

Intervention is triggered by deviations, safety alerts, obstacles, or requests.

# Supervised Autonomy Model

Full autonomy is deferred in favor of a practical model where one barber oversees multiple robots.



The background is a dark, deep space scene. In the upper left, a portion of a planet with rings is visible. In the lower right, a large, bright, curved horizon of a planet dominates the frame. The background is filled with numerous small, distant stars.

06

# Empowering Technologies

# Empowering Tech: Drive & AI



## Direct Drive Technology

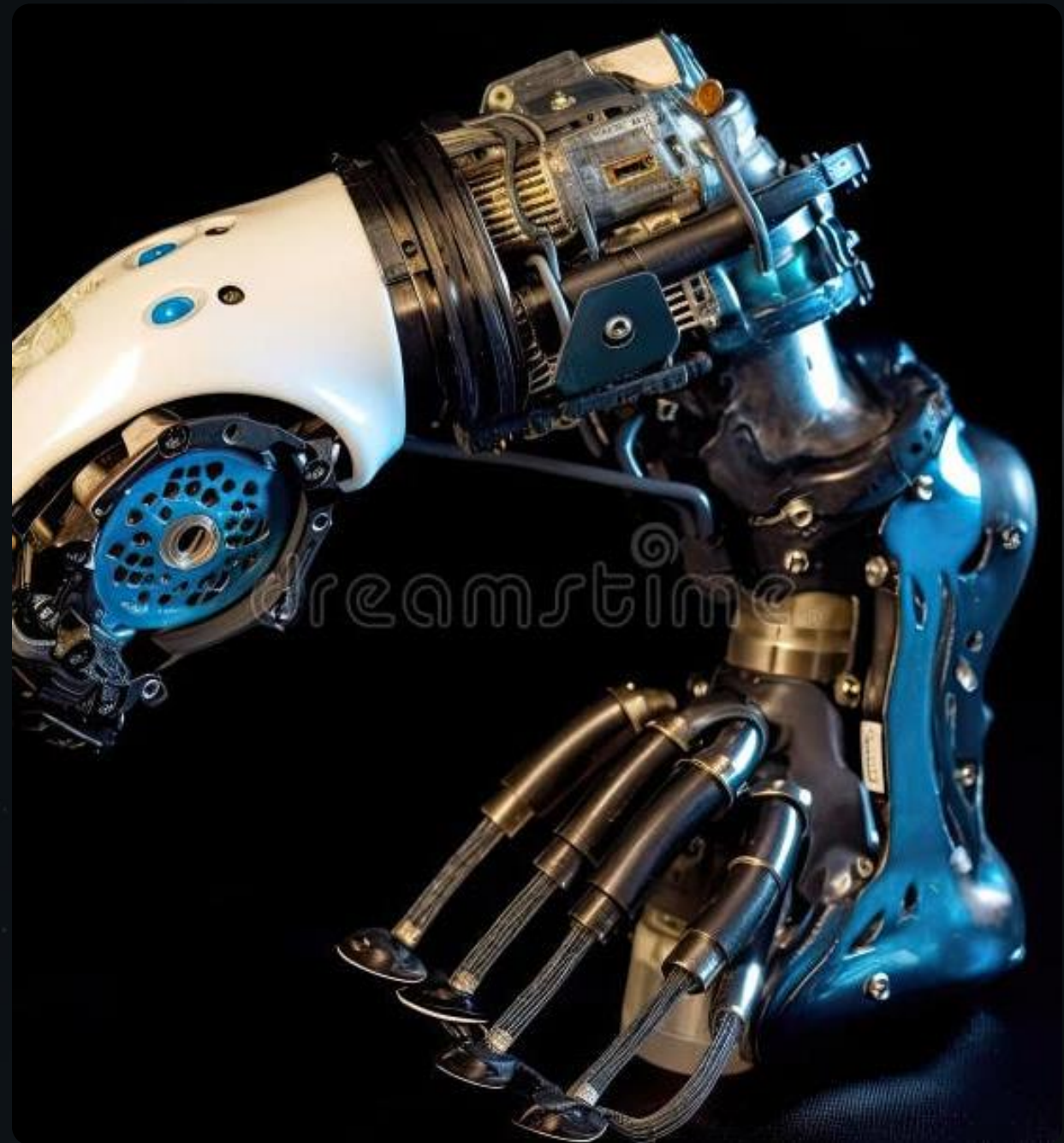
Provides precise, backlash-free motion crucial for intricate cuts and real-time head movement compensation.



## Large Language Models (AI)

Offer conversational companionship and personalized style advice, enhancing the customer experience.

Ref. S. Li, Book: Haircutting Robots, Springer.



# Empowering Tech: Immersion & Touch



## Virtual Reality (VR)

Immerses remote barbers in the client's environment for precise assessment and control.



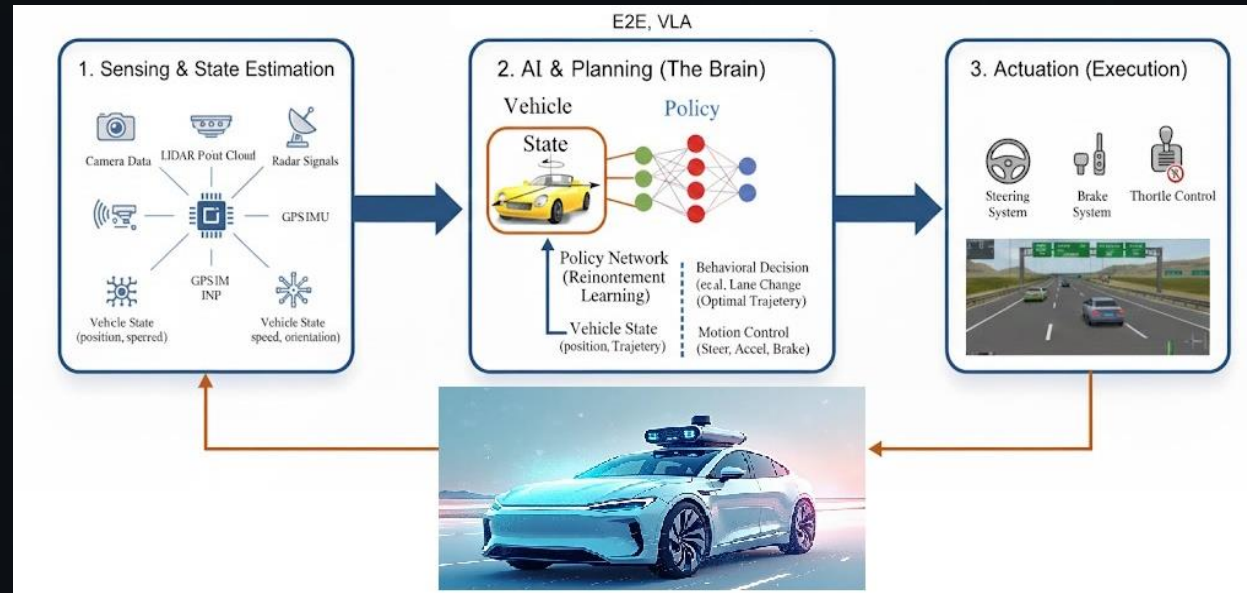
## Force Control and Haptic Feedback

Allows operators to feel cutting forces, enabling fine control and a safer experience.

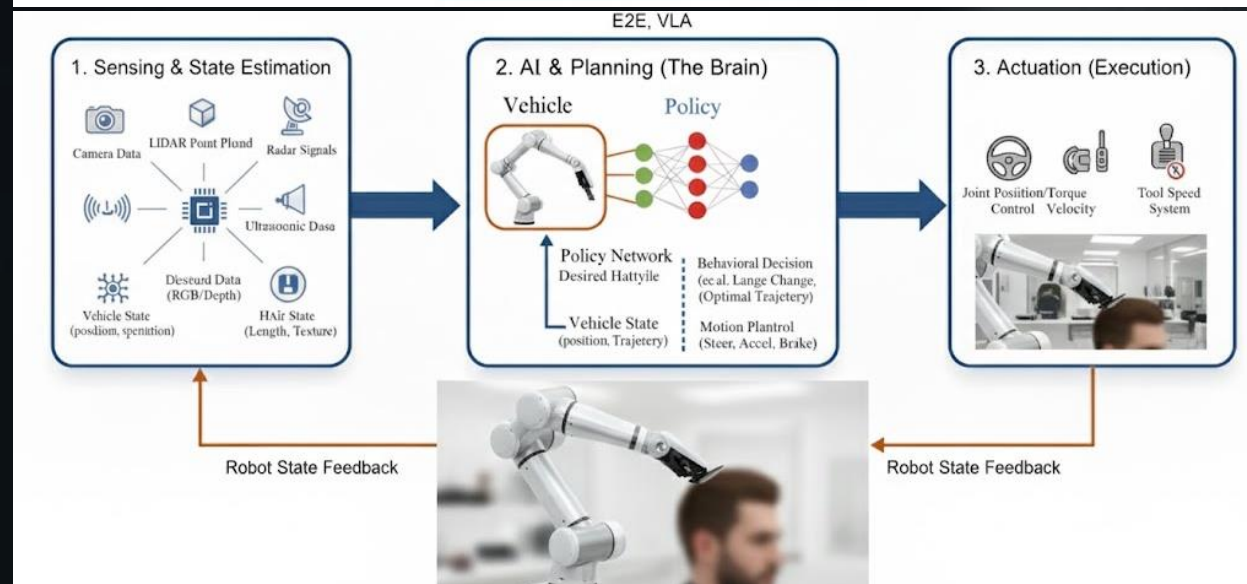
Ref. S. Li, Book: Haircutting Robots, Springer.

# Empowering Tech: E2E, VLA, Imitation Learning

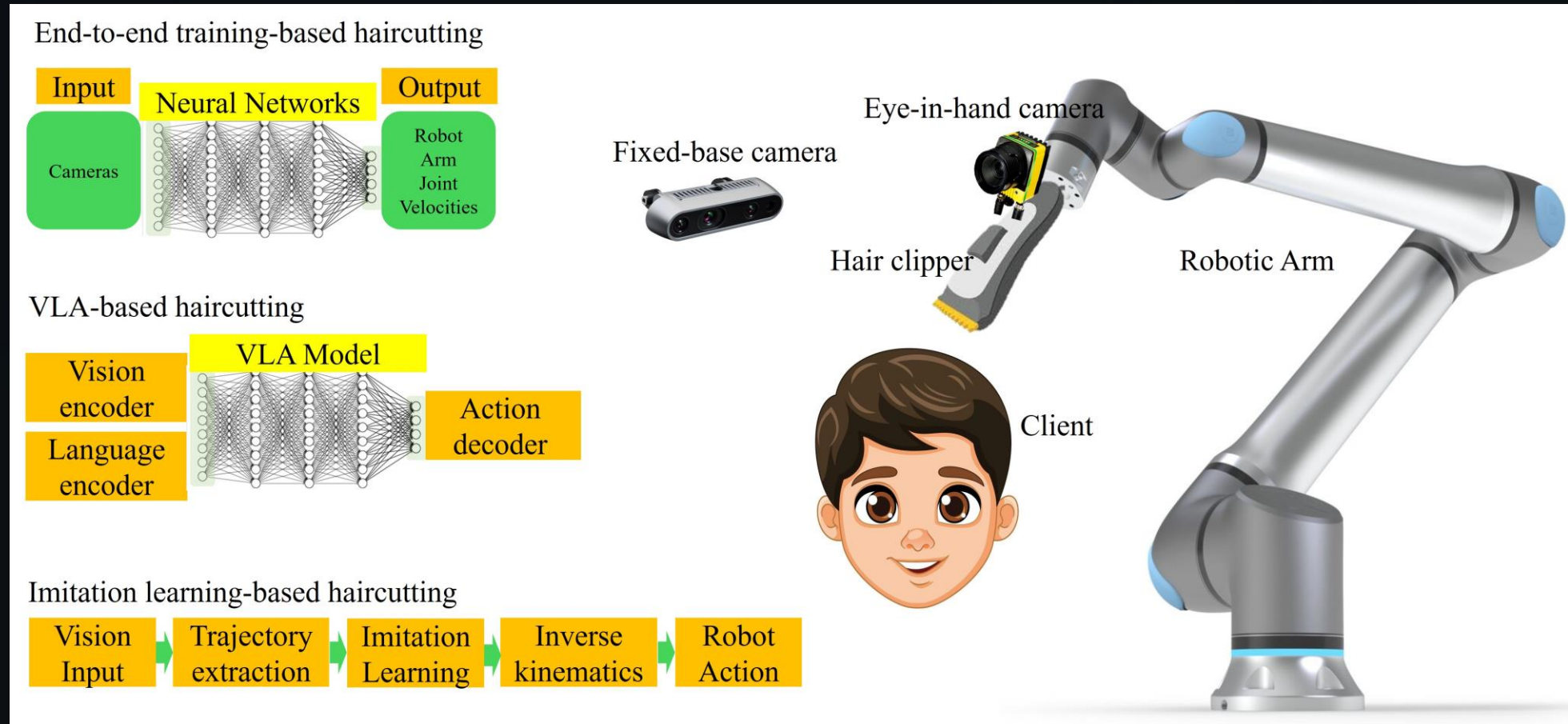
## Autonomous Driving



## Autonomous Haircutting

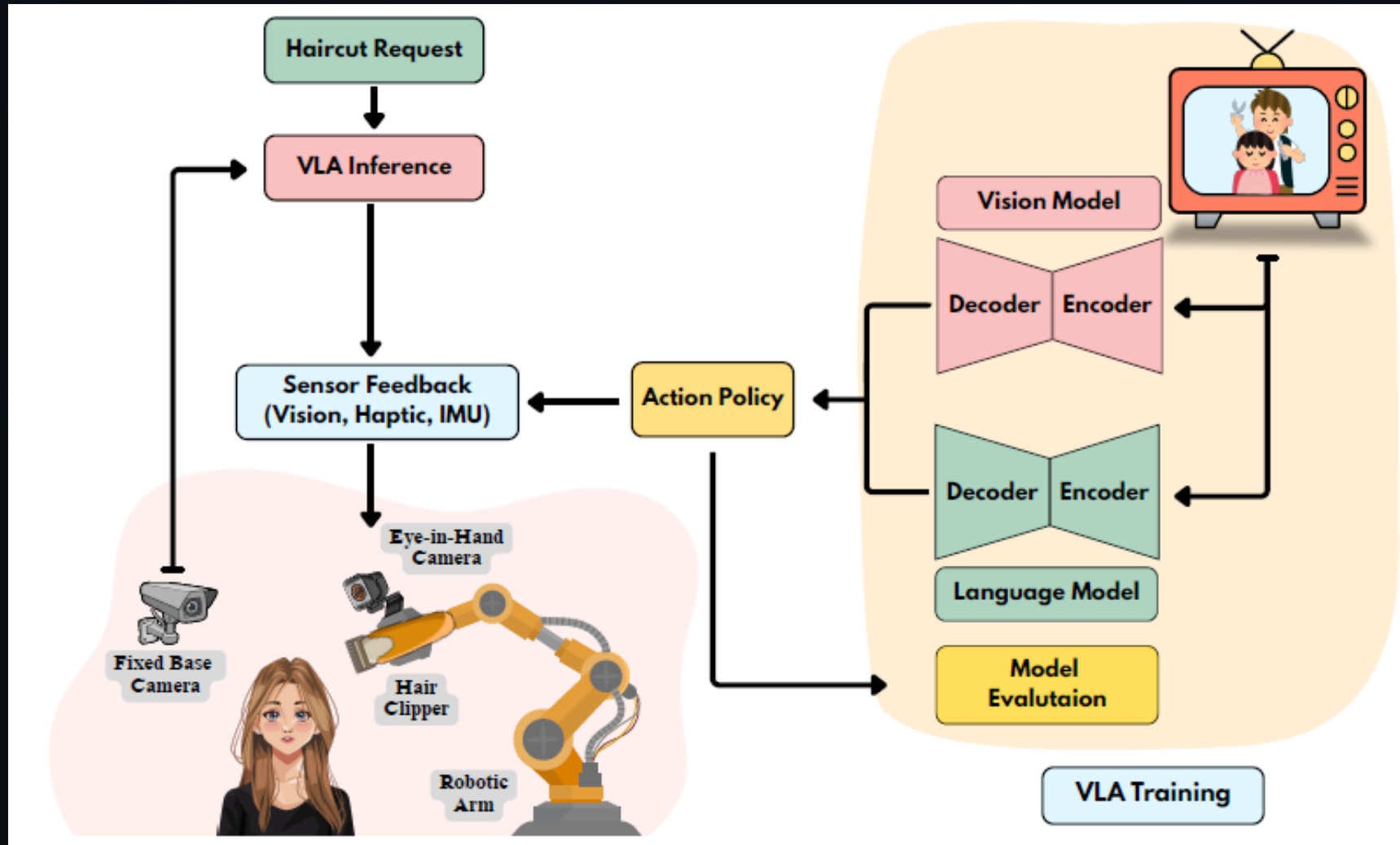


# Empowering Tech: E2E, VLA, Imitation



Ref: A. Khan and S. Li, Vision-Language-Action Models for Intelligent Haircutting Robots: A Position Paper on Architectures, Evaluation, and Future Directions, Submitted.

# Empowering Tech: E2E, VLA, Imitation



Ref: A. Khan and S. Li, Vision-Language-Action Models for Intelligent Haircutting Robots: A Position Paper on Architectures, Evaluation, and Future Directions, Submitted.

# Supporting Technology for Haircutting Robots



## Perception Layer

- 3D Visual Modeling (Structured Light)
- Force/Tactile Sensing (Anti-scratch)
- Hair Feature Recognition (Hardness/Density)



## Decision & Algorithm

- AI Hairstyle Design (Face Shape Matching)
- Motion Path Planning (Obstacle Avoidance)
- Intelligent Control (PID, Deep Learning)



## Execution Layer

- Multi-DOF Robotic Arm (6/7-axis)
- End Effector (Electric Clipper, Scissors)
- Direct Drive Servo (High Precision)



## HCI Layer

- Voice/Gesture Control Interaction
- AR Hairstyle Preview & Try-on
- Remote Control & Real-time Monitoring



## Safety & Security

- Collision Emergency Stop & Force Limit
- Power/Network Failure Safe Return
- Fault Detection & Predictive Maintenance

This five-layer closed-loop system integrates interdisciplinary technologies, enabling fully automated and safe haircut services.

# Perception Layer: Environment & Human Understanding

Core Objective: Achieve precise recognition and understanding of the head, hairstyle, and environment.



## 3D Visual Modeling

**Tech:** Structured Light, ToF, LiDAR

**Func:** Reconstruct head model, extract hair contours/lengths for path planning.

**Challenges:** Real-time dynamic modeling, lighting robustness.



## Force & Tactile Sensing

**Tech:** 6-axis Force Sensors, Tactile Arrays

**Func:** Detect contact force, enable feedback control for safety & comfort.

**Challenges:** Sensor durability, real-time signal processing.



## Hair Feature Perception

**Tech:** High-speed Cameras, Spectral Analysis

**Func:** Identify hardness, curliness, density for personalized cutting.

**Challenges:** Non-contact, fast & accurate analysis.

**Summary:** The Perception Layer acts as the robot's 'eyes' and 'skin'. It integrates multi-sensor data to create a precise digital twin of the user's head, enabling the system to make intelligent, safe, and personalized decisions during the haircut process.

# Decision & Algorithm Layer: The Intelligent Brain

**Core Objective:** Intelligently plan and control the entire haircut process based on perception data.



## AI Hairstyle Design

**Tech:** CV face analysis, golden ratio models, hairstyle DB.

**Func:** Recommends style via AR/VR preview; generates digital target model.

**Chall:** Quantifying aesthetics, satisfying personalized needs.



## Motion & Path Planning

**Tech:** Inverse kinematics, RRT\*, trajectory optimization.

**Func:** Plans collision-free paths for arm to reach target positions accurately.

**Chall:** High-redundancy path optimization, dynamic obstacle avoidance.



## Intelligent Control

**Tech:** PID, adaptive control, sliding mode control, RL.

**Func:** Precisely controls force/speed; learns to optimize cutting strategies.

**Chall:** High-dynamic real-time response, robustness in complexity.

---

This layer acts as the intellectual core, translating visual data into actionable, precise robotic movements through advanced computational intelligence.

# Execution Layer: Mechanics & Haircut Operation

**Core Objective: Precisely and stably execute haircutting actions**



## Mechanical Body Design

**Tech:** 6/7-axis robotic arms, lightweight materials, high-precision reducers.

**Func:** Full head coverage, obstacle avoidance, compliant operation.

**Chall:** Balancing precision, payload, and compactness.



## End Effector (Tool)

**Tech:** Integrated clippers, scissors, combs with auto-adjustment.

**Func:** Trimming, thinning, styling; anti-hair-pinching safety.

**Chall:** Miniaturization, reliability, and quick tool change.



## Drive System

**Tech:** High-precision servo motors, direct-drive, harmonic drives.

**Func:** Power movement, ensure precise control & fast response.

**Chall:** System efficiency, low noise, and long-term stability.

Translating decisions into precise physical movements with industrial-grade accuracy.

# Human-Computer Interaction Layer

*Core Objective: Achieve a convenient, intuitive, and user-friendly interaction experience.*



## Interaction Methods

**Technologies:** Speech recognition, computer vision gesture recognition, HD touch screens.

**Function:** Control robot via voice, gestures or touch. NLP enables fluent, humanized interaction.

**Challenges:** Accuracy in complex environments, gesture robustness, logic simplicity.



## Visual Interaction

**Technologies:** AR/VR, 3D rendering engines.

**Function:** Visualize virtual hairstyles on user's head pre-cut. Display real-time progress.

**Challenges:** Accurate virtual-real matching, real-time rendering efficiency.



## Remote Interaction

**Technologies:** IoT, 5G communication, remote control software.

**Function:** Remote booking, status monitoring, and expert remote guidance for complex styles.

**Challenges:** Low-latency control, network security & privacy.

---

The Human-Computer Interaction Layer bridges the gap between user intent and robotic execution. By integrating advanced perception and visualization technologies, we aim to make the haircutting process not just automated, but truly intelligent and empathetic.

# Safety & Security Layer: Bottom-line Guarantee

Core Objective: Ensure absolute safety during the haircut and stable and reliable system operation.



## Safety Mechanisms

**Technologies:** Emergency stop, collision detection, force feedback, safety light curtains.

**Function:** Multi-redundancy system. Immediate stop on abnormal force or collision. Electrical safety design.

**Challenges:** Fast response, avoiding false triggers.



## Reliability Design

**Technologies:** Fault diagnosis, sensor redundancy, modular design, thermal management.

**Function:** Real-time monitoring, failure prediction, proactive maintenance, high availability.

**Challenges:** Complex system failure analysis, predictive accuracy.



## Compliance & Ethics

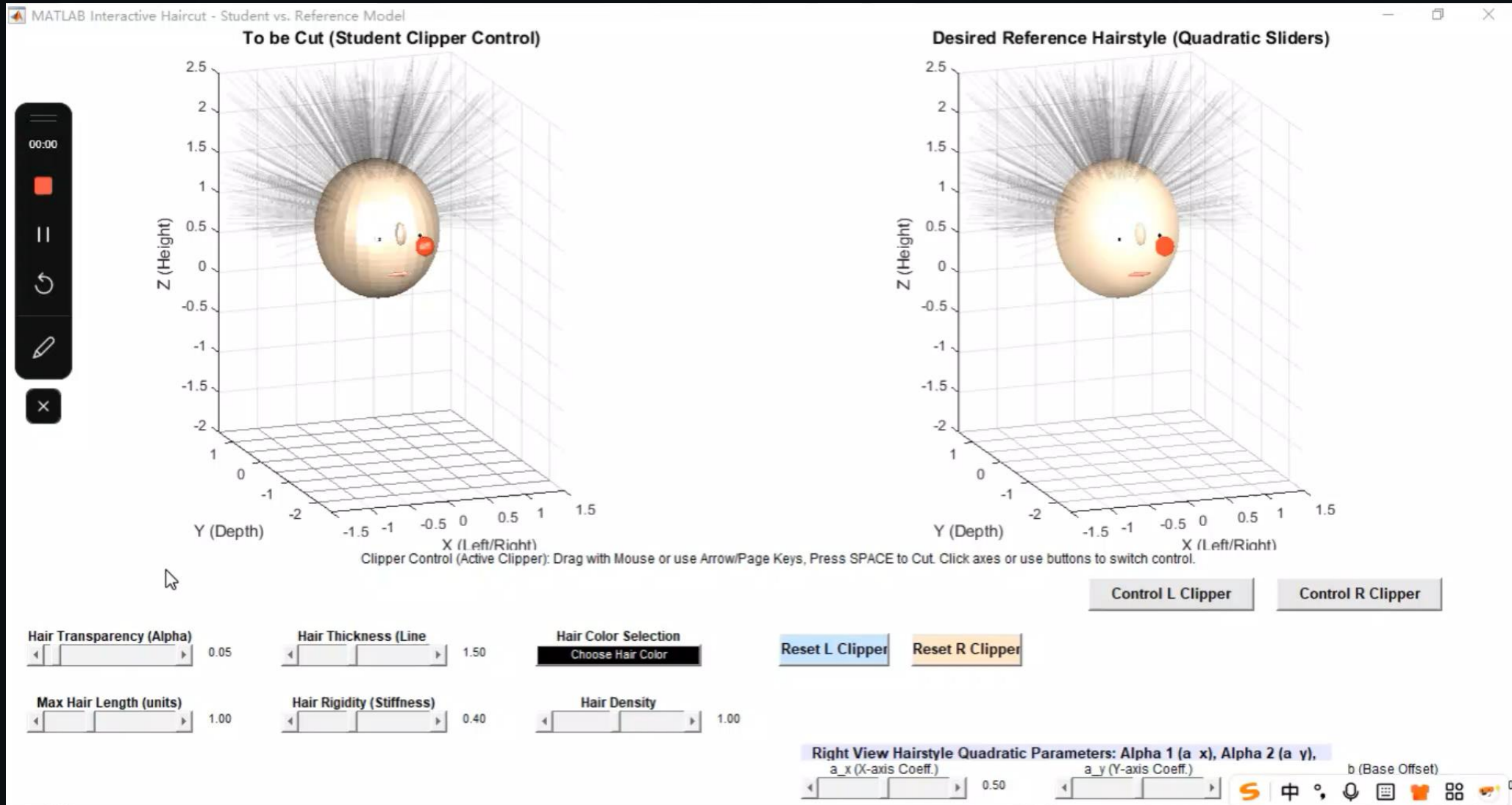
**Technologies:** Data encryption, access control, privacy protocols.

**Function:** Strict protection of facial/hairstyle data. Compliance with safety standards and norms.

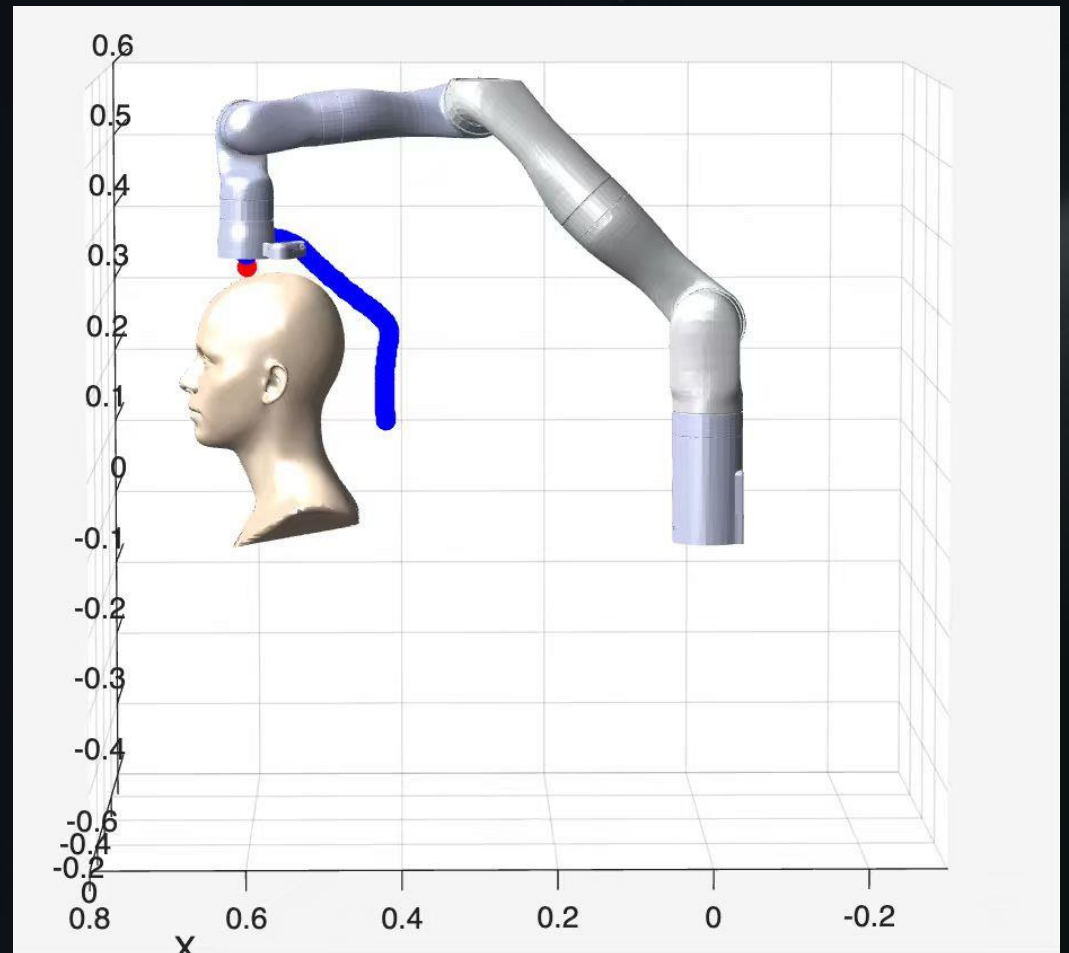
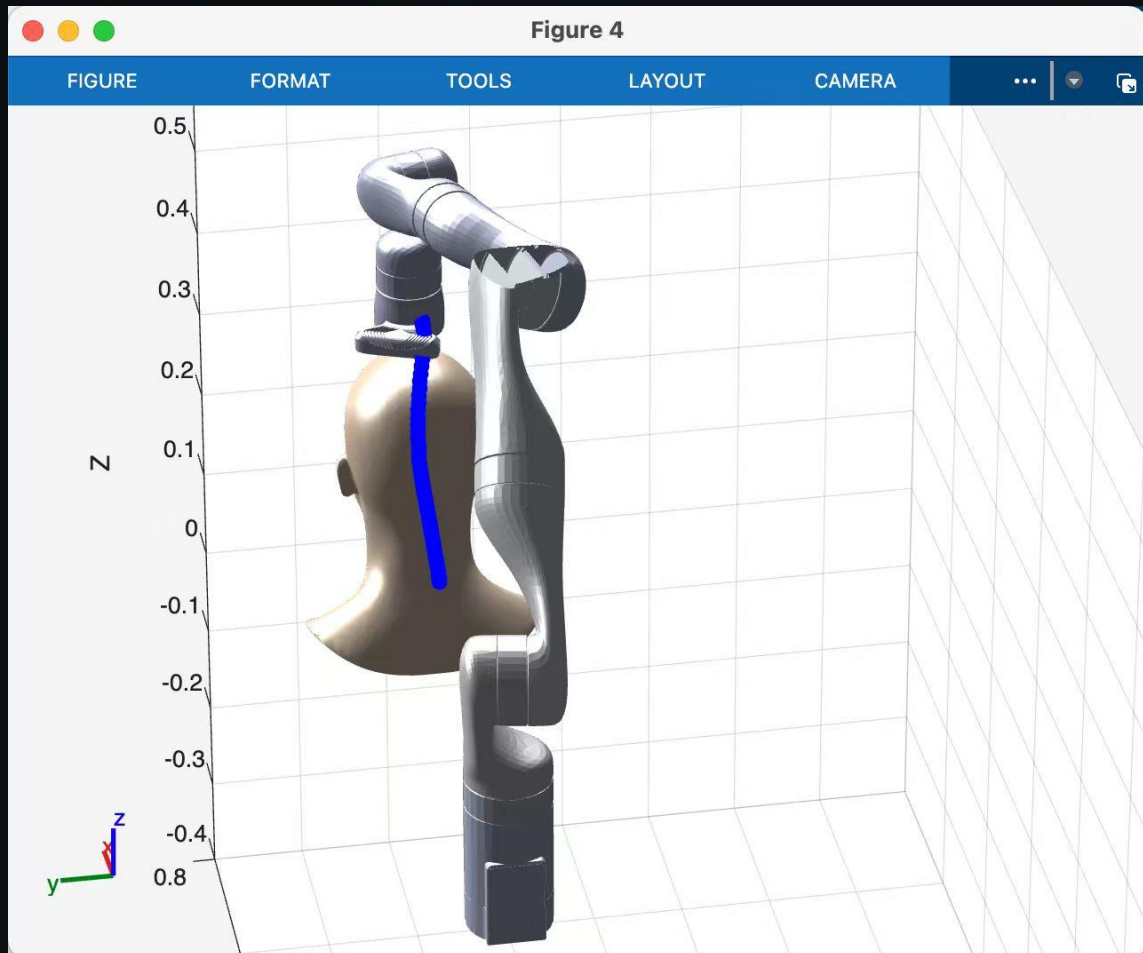
**Challenges:** Balancing data utilization with privacy, adapting to regulations.

This layer serves as the foundation for user trust, ensuring that innovation never compromises on safety or privacy.

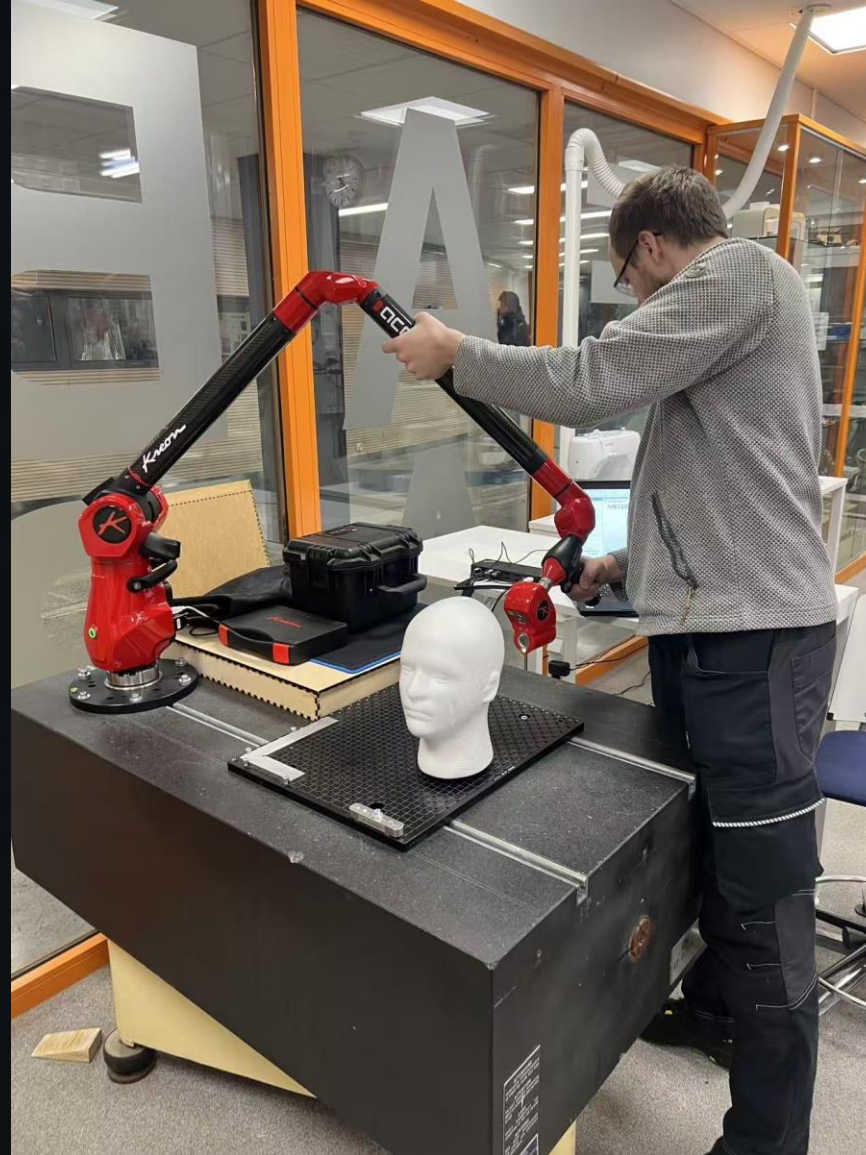
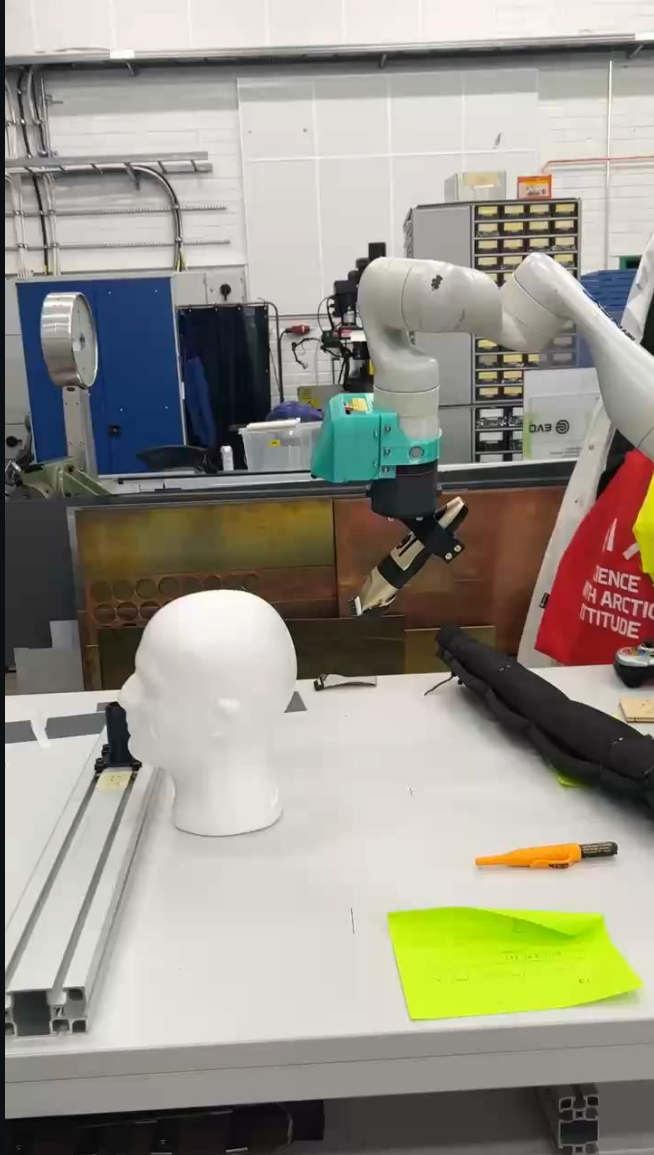
# Preliminary Results: Hair Dynamics Simulator



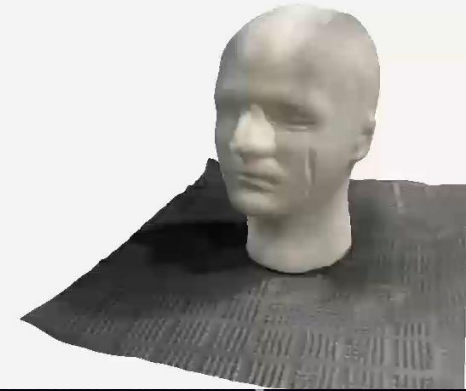
# Preliminary Results: Haircutting Robot Simulator



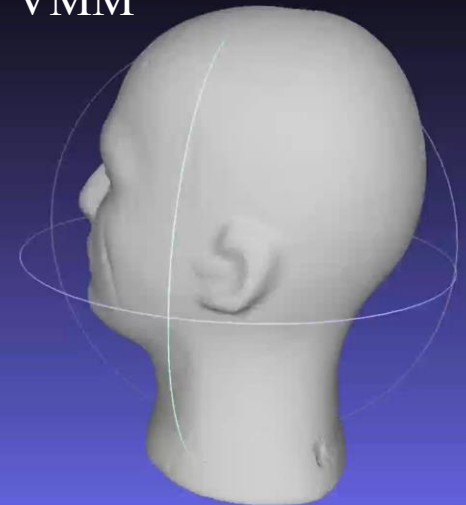
# Preliminary Results: Experimental Systems



By Iphone Lidar

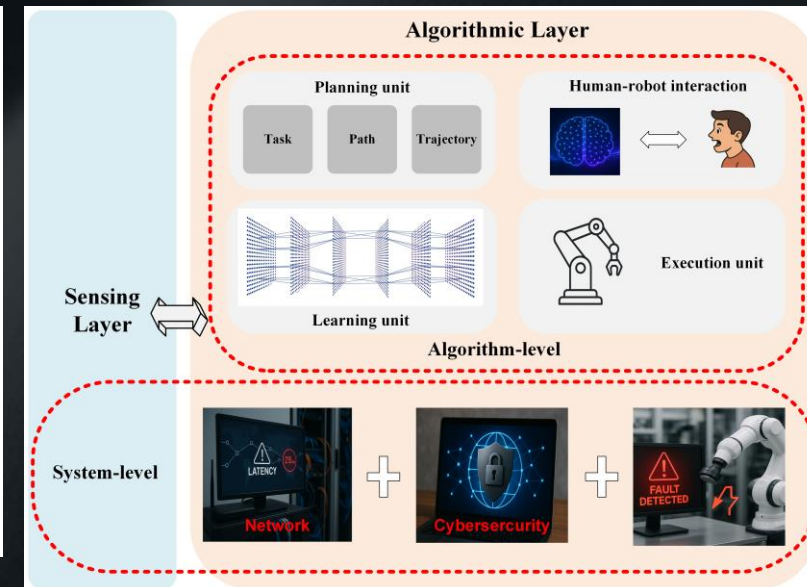
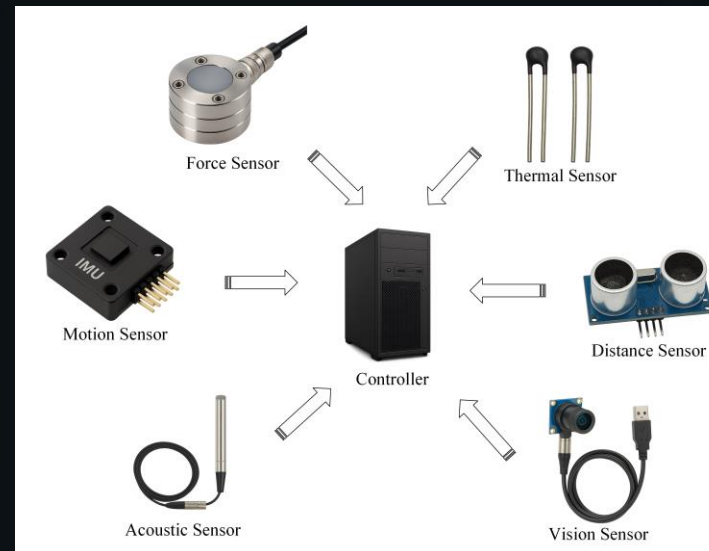
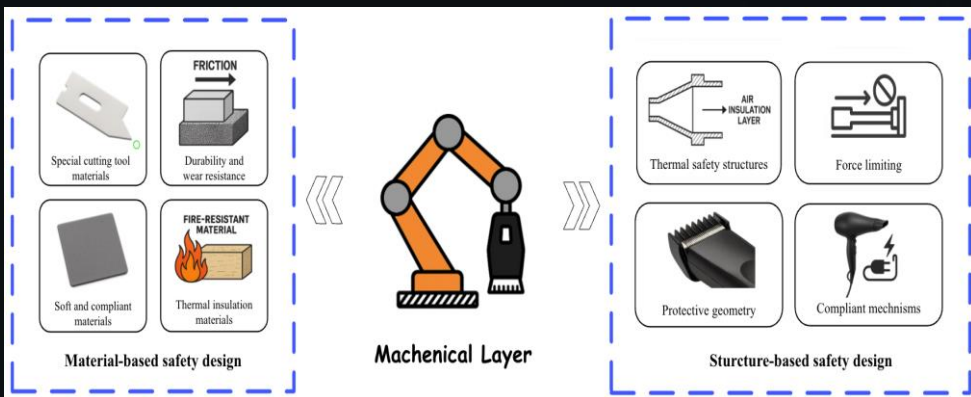
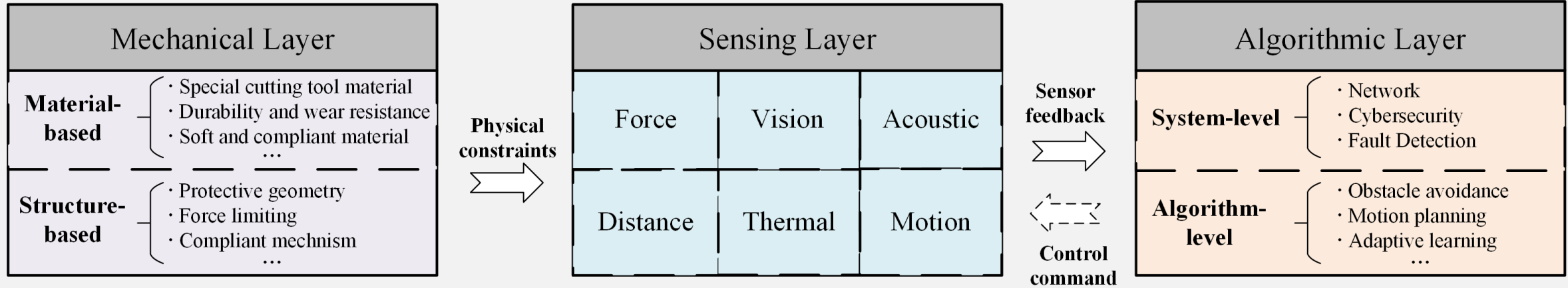


By VMM



# Preliminary Results: Safety Mechanisms

## Haircutting Robot



# Preliminary Results: Safety Mechanisms

*Mechanical Layer*



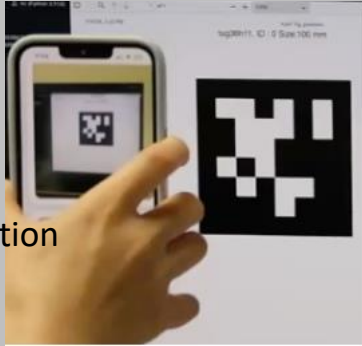
*Hair Clipper*

Z. Huang, A. Vilkkki, S. Li and J. Röning, Safety in Robotic Haircutting, Under 2<sup>nd</sup> round of review in Journal of Field Robotics

# Preliminary Results: Teleoperation

## Master side

AprilTag pose estimation



3D vision reconstruction

pose signal

control command



Binocular video

Binocular video

## Slave side

UR10e robotic arm

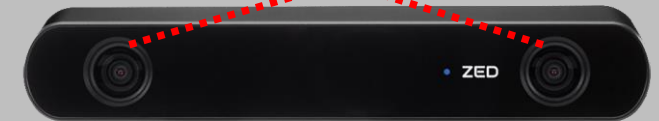


## Vision side



Left-eye image

Right-eye image



ZED stereo camera

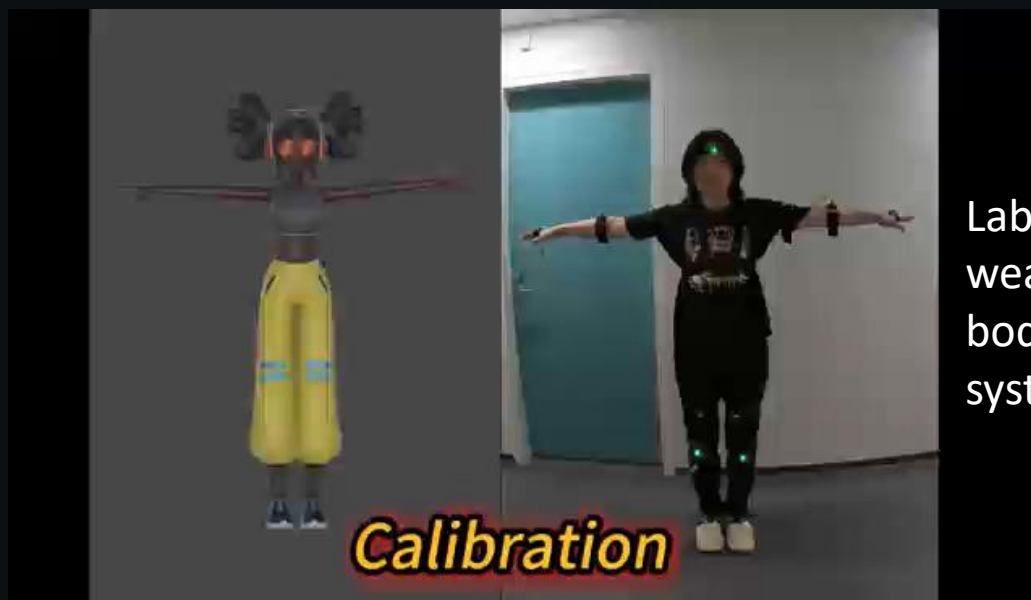
# Preliminary Results: Teleoperation



Lab-made  
wearable  
body tracking  
system for  
dual arm  
teleoperation



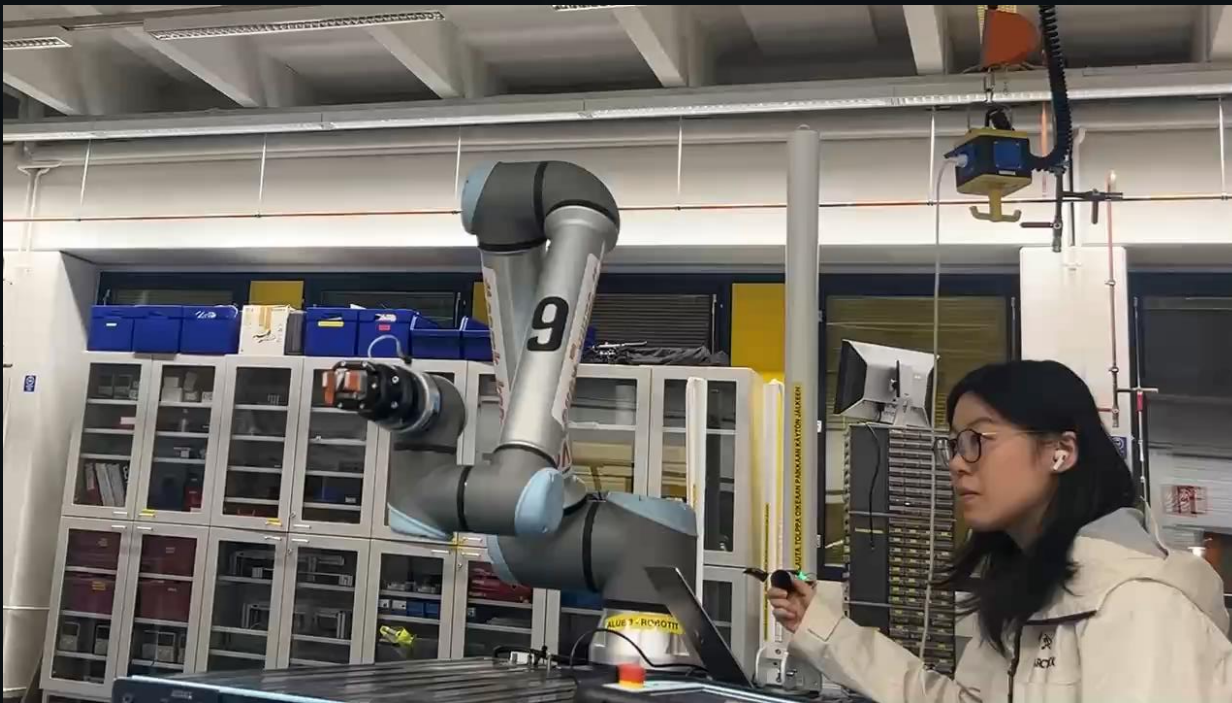
Lab-made  
wearable  
body tracking  
system for  
teleoperat  
ed  
haircutting



Lab-made  
wearable  
body tracking  
system

# Preliminary Results: Teleoperation

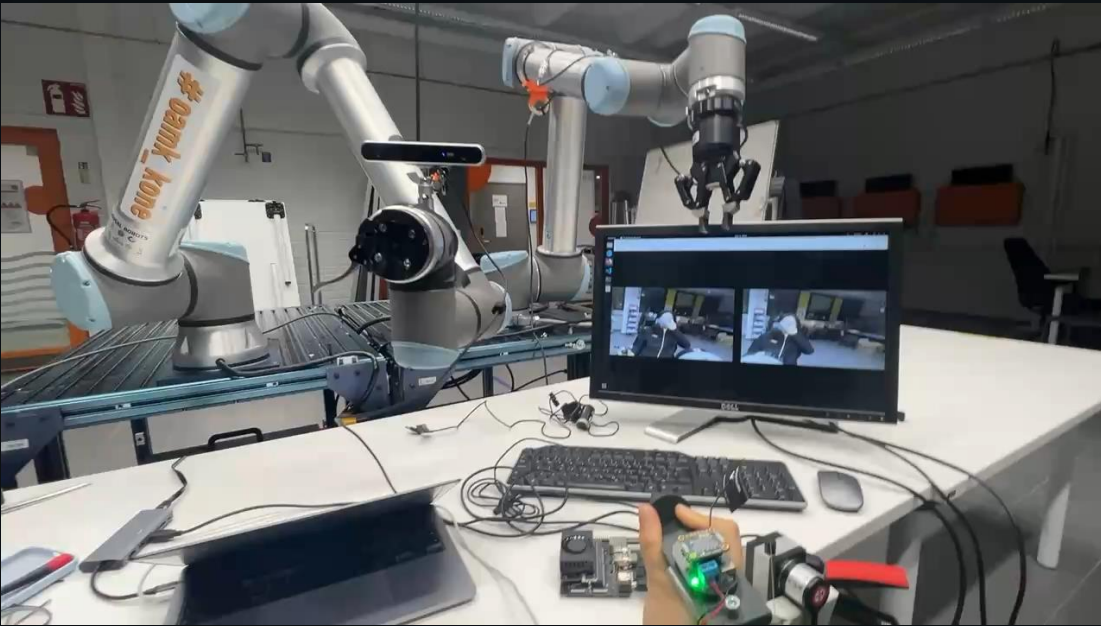
Lab-made control handle for teleoperated haircutting



Lab-made control handle (Cable CMM) for 6 DoF positioning (Ref. Z. Huang, Z. Chen, A. T. Khan, Y. Gan, F. Hu, J. Röning, and S. Li, A hybrid 6-DoF pose measurement system using string potentiometer and dual IMUs, 2025 IEEE International Workshop on Metrology for Sustainability, 2025)

# Preliminary Results: Teleoperation

Experiments of Teleoperation in our lab.



Telehairing: an open accessible teleoperated  
haircutting robot

Experiments of Teleoperation in our lab.

Ref.: Z. Huang, S. Li, TeleHairing: A Teleoperation Baseline for Robotic Haircutting, submitted.

# Preliminary Results: Teleoperation



Lab-made 6 DoF robotic arm using Xiaomi cybergear motors (Z. Chen, Z. Huang, Y. Gan, F. Hu, J. Röning, and S. Li, Development of a 6-DoF Robotic Manipulator with Smart Actuators and Real-Time Inverse Kinematics, ICNSC 2025.

Manipulability Optimization Control of a Serial Redundant Robot for Robot-assisted Minimally Invasive Surgery

Robotic surgery using self developed control algorithms (Ref. H. Su, S. Li, L. Bascetta, G. Ferrigno and Elena. Momi, Manipulability Optimization Control of a Serial Redundant Robot for Robot-assisted Minimally Invasive Surgery, ICRA 2019.)

# Preliminary Results: Teleoperation

## Normal Operation



## Rebase Operation



## Safety Governor



## Emergency Stop



Ref. Z. Huang, A. Khan, A. Mohammed, F. Regio, J. Torvinen, L. Angrisani and S. Li, A Safety Boundary Governor for Teleoperated Robotic Haircutting, submitted to IEEE MetroSustainability 2026

The background is a dark, cosmic scene. In the upper left, a portion of a planet with rings is visible. In the lower right, a large, bright, curved horizon of a planet dominates the frame. The background is filled with numerous small, distant stars.

07

# Values & Significances

# Values & Significances

*Haircutting robotics reconstructs the beauty industry's value chain by establishing a human-machine collaboration model where automation handles standardized repetitive labor, thereby democratizing service access, protecting occupational health, elevating creative value, and ensuring inclusive accessibility for marginalized groups.*

## 01. TARGET

Focuses on tedious, repetitive standard haircuts while reserving high-end artistic styling for human barbers, establishing clear human-machine collaboration boundaries.

## 02. AFFORDABILITY

Drives down operational costs, making consistent, basic haircut affordable and accessible to a wider demographic, democratizing personal grooming.

## 03. OCCUPATIONAL HEALTH

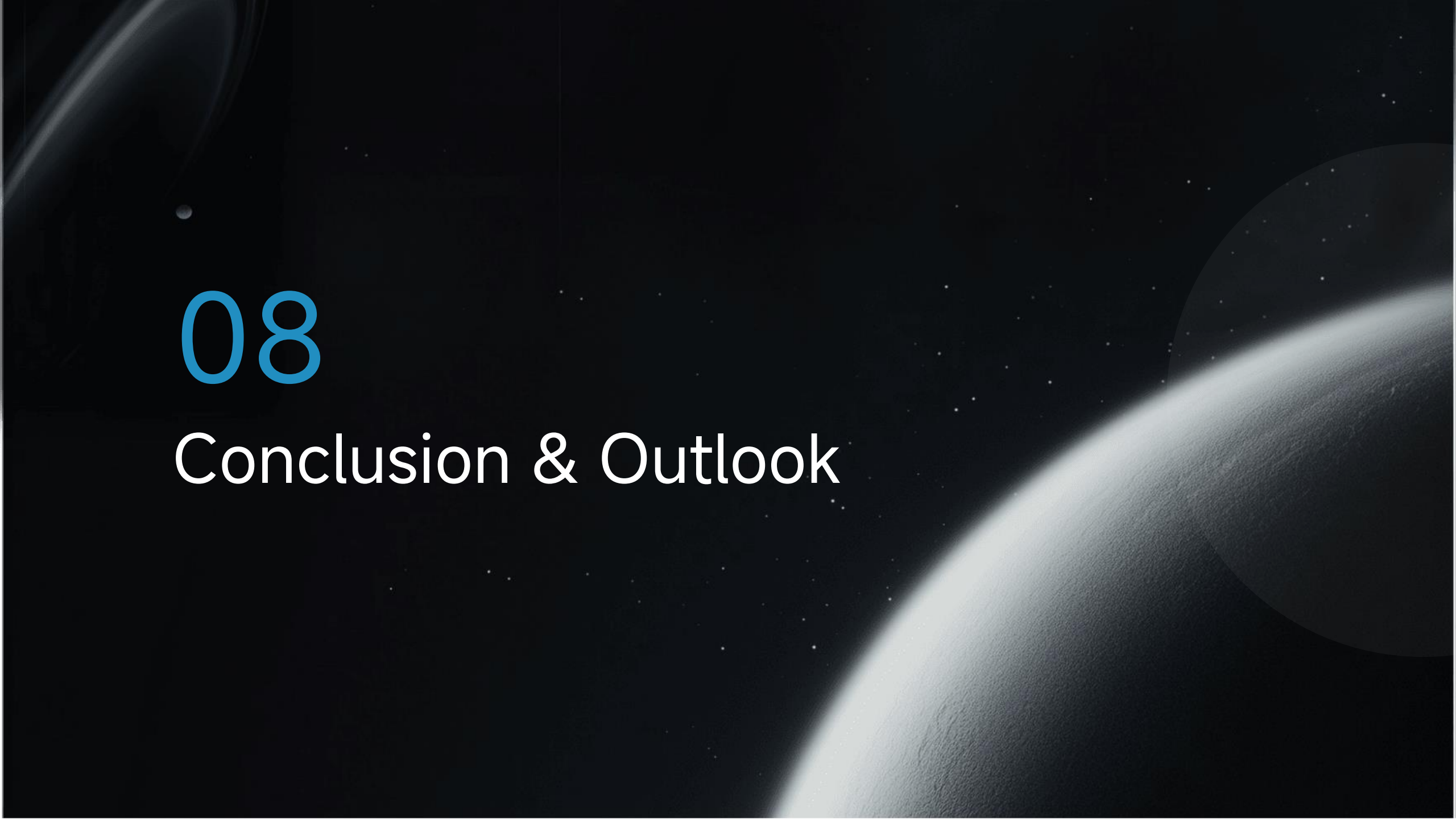
Eliminates repetitive physical strain for stylists, drastically reducing occupational risks like chronic back pain and muscle fatigue from long hours.

## 04. VALUE UPGRADE

Frees barbers from boring, standardized repetitive tasks to focus on creative, artistic expression and client consultation. Shifts the profession from standardized labor to high-value, human-centric creativity and design.

## 05. INCLUSIVITY

Enables independent haircare access for people with mobility challenges or social anxiety through future in-home robotic solutions, fostering greater equity and dignity in personal care.

A dark space background featuring a large, bright, curved horizon of a planet or moon in the lower right. In the upper left, a ringed planet like Saturn is visible. The background is filled with numerous small, distant stars.

08

## Conclusion & Outlook

# Key Takeaways



## Convergence of Tech

Cheaper robots, rising labor costs, and advanced AI are creating a perfect storm for automation.



## Practical Deployment

Supervised autonomy and robust safety engineering are the keys to early commercialization.



## Massive Market

A multi-faceted digital platform approach unlocks a global market worth hundreds of billions.

# References

- S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation, 2025.
- S. Li, Book: Haircutting Robots, Springer, 2025.
- Khan and S. Li Robotic Haircutting Systems: A Survey of Methods, Challenges, and Hair Modeling Insights, IEEE Journal of Selected Areas in Sensors, 2026.
- A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation, 2026.
- Z. Chen, Z. Huang, Y. Gan, F. Hu, J. Röning, and S. Li, Development of a 6-DoF Robotic Manipulator with Smart Actuators and Real-Time Inverse Kinematics, ICNSC 2025
- H. Su, S. Li, L. Bascetta, G. Ferrigno and Elena. Momi, Manipulability Optimization Control of a Serial Redundant Robot for Robot-assisted Minimally Invasive Surgery, ICRA 2019.
- Z. Huang, Z. Chen, A. T. Khan, Y. Gan, F. Hu, J. Röning, and S. Li, A hybrid 6-DoF pose measurement system using string potentiometer and dual IMUs, 2025 IEEE International Workshop on Metrology for Sustainability, 2025
- Z. Huang, A. Vilkki, S. Li and J. Röning, Safety in Robotic Haircutting, Journal of Field Robotics
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- Z. Huang, A. Khan and S. Li, Haircutting Robots: A Mobile Robotics Perspective, International Journal of System Sciences.
- Z. Huang, S. Li, TeleHairing: A Teleoperation Baseline for Robotic Haircutting, submitted. 2026.
- Z. Huang, A. Khan, A. Mohammed, F. Regio, J. Torvinen, L. Angrisani and S. Li, A Safety Boundary Governor for Teleoperated Robotic Haircutting, submitted to IEEE MetroSustainability, submitted, 2026
- Online Repository: <https://github.com/Dai0731csc/Robotic-Haircutting-Publications>



Questions or Comments

Thanks!